

Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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Motivation

- **Economic Problem:** Selecting best options under costly evaluation
 - Determining each option's true quality costs decision-maker time and effort
 - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
 - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
 - Graduate programs invest time and money into students' development
 - If programs are undervaluing applicants with certain characteristics, future such applicants may receive currently-withheld opportunities

Research Questions

- ① Are admissions committees optimizing, or are they making mistakes?
 - Goal is to accept applicants with highest success probabilities who will matriculate
 - Per literature, measure success with program completion and job placement ranking
 - Predictors of acceptance only (success only) may be overweighted (underweighted)
 - ② If they are making mistakes, how can they make better admissions decisions?
 - In turn, by changing decision rules, how much better of a cohort can they admit?
- ★ Methods apply to undergraduate/professional admissions, hiring, and other settings

Methodology and Main Results

- Build structural models of admissions committee's decision problem that support:
 - Statistical tests of whether committee's decision rule aligns with desired objective
 - Counterfactual simulations quantifying gains from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
 - For most applicants, estimated success probabilities align with those implied by committee's decisions. But some applicant characteristics are misweighted
 - Correctly weighting these characteristics would increase admitted cohort's average success probability by about 4 percentage points without reading more files
 - Estimate corrected for winner's curse $\Rightarrow$ reflects out-of-sample performance

Contribution to Literature

- **Graduate Admissions:** Regress acceptance and success on applicant characteristics
 - Krueger & Wu (2000); Athey, Katz, Krueger, Levitt & Poterba (2007); Stock, Finegan & Siegfried (2006–2015); Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
 - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
 - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)
 - **Optimality:** Autor & Scarborough (2008); Cho & Rust (2010); Kleinberg et al. (2018)
- ★ **This Paper:** Builds and estimates dynamic selection models of admissions, which test optimality by relating acceptance effects to success effects

Synthetic Data Protocol (IRB-Approved)

① **Administrative office receives data from university departments**

- Algorithmically extracts variables from textual files, such as recommendation letters

② **Synthetic data (Patki et al. 2016) used for model development**

- Estimate joint distribution of variables, and simulate synthetic dataset from it
- Rows do not correspond to real subjects, preventing identity disclosure
- Cells with < 5 comparable observations binned to prevent attribute disclosure

③ **Administrative office executes final estimation code on actual data**

- Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

Outline

- 1 Regression Analysis
- 2 Structural Models
- 3 Structural Estimation
- 4 Optimality and Gains

Data Description

- **Sample:** Applicants to research university's economics PhD program from 2015–2019
- **Outcomes:** Measures of admission and success
 - **Admission:** Read File $\Rightarrow$ Accept $\Rightarrow$ Matriculate $\Rightarrow$ Success
 - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Covariates:** Objective vs. subjective variables
 - **Objective:** Demographics and performance measures, such as GRE scores
 - **Subjective:** Textual processing of recommendation letters, statements of purpose, CVs, and supplemental forms using dictionaries to construct variables

Regression Results

- **Method:** Logistic regressions for acceptance, matriculation, and success
 - **Success:** *Top 100 Placement* = 1 for 20% of applicants vs. 34% of matriculants
- **Results:** Different independent variables best predict each dependent variable
 - **Accept:** *Committee Score* is best predictor. Other predictors: demographics, *GRE Quantitative Percentile*, *Work Experience*, *CV Coding* (negative)
 - **Matriculate:** Predictors of acceptance often negatively predict matriculation. Best predictor of matriculation is *Open House* attendance
 - **Success:** *PhD Program Rank* is best predictor. Other predictors: *Age*, *TOEFL/IELTS*, *Work Experience*, *Essay Positive Terms (%)*

Acceptance vs. Success AMEs

Dependent Variable	Accept	T100 Placement
Winsorized Age	-0.0001	0.0053*
Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

» Insufficiency of Comparing AMEs

» Diversity Objective

» Regressions vs. Model

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to test optimality
- Need model of admissions to relate acceptance effects to success effects

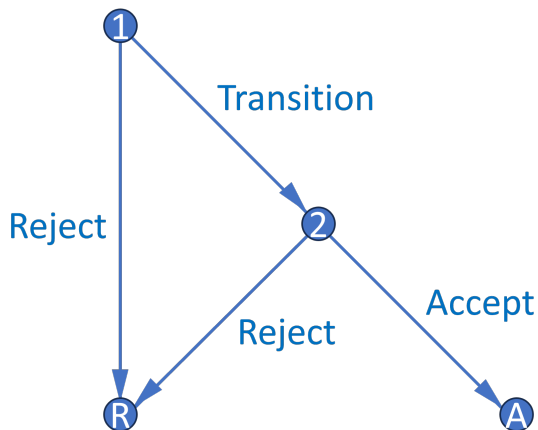
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- 1 Regression Analysis
- 2 Structural Models**
- 3 Structural Estimation
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Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
 - Per literature, committee should accept applicants with highest success probabilities
 - Committees also accept applicants with higher matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
 - Quantify gains from deviating from actual to counterfactual decision rule

Dynamics



- **Round 1:** See only objective variables. Transition to Round 2 or reject
- **Round 2:** See survivors' objective and subjective variables. Accept or reject
- Assume for now number of applicants transitioned and accepted is fixed
- Matriculation model will account for success and matriculation probabilities

Portfolio Choice Problem

- Accept applicants with highest success probabilities under capacity constraints
- For applicant n , choose $d_{1,n} \in \{\text{Transition, Reject}\}$ and $d_{2,n} \in \{\text{Accept, Reject}\}$:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N_T} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_A,$$

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N \underbrace{EV_{2,n}(x_n; \bar{\lambda}_2^*)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

where x = objective variables, z = subjective variables, $y \in \{\text{Success, Failure}\}$,
 N = number of applicants, (N_T, N_A) = max number of files read and acceptances

Portfolio Choice Problem

$$\begin{aligned} \max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} & \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N_T} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_A \\ \max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} & \sum_{n=1}^N \underbrace{EV_{2,n}(x_n; \bar{\lambda}_2^*)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T, \end{aligned}$$

- Portfolio choice problem has cutoff rule solution
 - **Round 2:** Accept transitioned applicant if success probability exceeds Round 2 cutoff
 - **Round 1:** Transition applicant if Round 2 expected value exceeds Round 1 cutoff
 - Decisions across applicants independent, so no need to consider all subsets

Cutoff Rule Model

- ★ **Goal:** Use choices and outcomes data to estimate restricted success probabilities, while respecting constraint on number of files committee can read
- ★ **Legend:** Estimated model objects vs. Choices and outcomes data
- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Cutoff Rule Model

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Cutoff Rule Model

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Cutoff Rule Model

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

Conditional Choice Probabilities:

$$P(d_1 | x, t, \bar{k}) = \frac{\exp(u_1(\cdot, d_1)/\sigma)}{\sum_{\tilde{d}_1 \in \{T, R\}} \exp(u_1(\cdot, \tilde{d}_1)/\sigma)}, \quad P(d_2 | x, z, t, \bar{k}) = \frac{\exp(u_2(\cdot, d_2)/\sigma)}{\sum_{\tilde{d}_2 \in \{A, R\}} \exp(u_2(\cdot, \tilde{d}_2)/\sigma)}$$

►► Dynamic Parameterization

►► Static Model

Portfolio Choice with Matriculation

- Accept applicants with highest success probabilities who are expected to matriculate
- For applicant $n \in \{1, \dots, N\}$, choose $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^N \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M,$$

where (x, z) = variables, $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$,
 N = number of applicants, N_M = max number of matriculants

Portfolio Choice with Matriculation

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

such that: $\sum_{n=1}^N \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M$

Theorem 1

*For large N , this problem has a cutoff rule solution on **success** probabilities.*

- Likely matriculants more valuable, but also more expensive $\Rightarrow$ effects cancel
- Matriculation probabilities matter in Round 1 of a two-round model

Cutoff Rule with Matriculation

- ★ **Goal:** Two-round cutoff rule model where matriculation matters
- ★ **Legend:** Estimated model objects vs. Choices and outcomes data
- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = cutoffs

Cutoff Rule with Matriculation

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = cutoffs

Cutoff Rule with Matriculation

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S, m_1^* = M | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = cutoffs

Cutoff Rule with Matriculation

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

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Importance of Matriculation:

$$\lim_{\sigma \rightarrow 0} EV_2 = \mathbb{E}_{\tilde{z}}[\max\{P(y = S | x, \tilde{z}, t, \bar{k}) - P(y_2^* = S | t), 0\} P(m = M | x, \tilde{z}, t) | x, t, \bar{k}]$$

- Matriculation matters when expected success probability $\gg$ Round 2 cutoff
- **Upshot:** Low-ranked programs must consider matriculation in Round 1

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Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^N \underbrace{f(z_n | x_n, t_n; \theta_F)}_{\text{Density of } z|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$
$$\prod_{n=1}^N \underbrace{P_{d_{1,n}}(\theta_T)^{\mathbb{1}(d_{1,n}=T)} (1 - P_{d_{1,n}}(\theta_T))^{\mathbb{1}(d_{1,n}=R)}}_{\text{Round 1 conditional choice probabilities}} \prod_{n=1}^{N_T} \underbrace{P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Round 2 conditional choice probabilities}}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with flexible logistic specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- $\mathcal{L}^R$ aligns success probabilities implied by outcomes with those implied by CCPs

Cutoff Probability Identification

- Identify each year's cutoffs by how selective committee is that year
 - More acceptances and stronger applicants increase cutoff
 - Fewer acceptances and weaker applicants decrease cutoff
- While selectivity identifies cutoffs, sorting improves restricted likelihood

Theorem 2

Cutoff probability estimate is such that number of acceptances N_A equals expected N_A . For any cutoff and N_A , likelihood is highest when committee accepts best N_A applicants.

Restricted Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.3731 (0.3067, 0.4539)	36.58% (29.81%, 43.92%)	
⋮	⋮	⋮	
2019	0.3887 (0.3268, 0.4624)	38.14% (31.89%, 44.81%)	
Sigma			0.0020 (0.0013)

- Round 1 cutoffs relative to EV_2 . Round 2 cutoffs relative to success probabilities
- Cutoffs stable across years. $\hat{\sigma}$ small $\Leftrightarrow$ distorted restricted success probabilities

Restricted Estimates with Matriculation

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.0071 (0.0026, 0.0194)	36.35% (29.47%, 43.84%)	
⋮	⋮	⋮	
2019	0.0095 (0.0035, 0.0255)	37.41% (31.04%, 44.26%)	
Sigma			0.0020** (0.0010)

- With matriculation, EV_2 's functional form yields lower Round 1 cutoffs

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Deviation Gains Test

- Suppose committee adopted model's decision rule. Would average success probability of acceptance/matriculation set have increased? If so, by how much?
 - Select applicants with model, compute average yearly success probability of model's acceptance/matriculation set, and compare to that of actual decision rule's set
 - Plug $\hat{\theta}_S^U$ into model because $\hat{\theta}_S^R$ may be distorted to rationalize suboptimal behavior
 - Compute average success probabilities across $B = 500$ perturbations $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$

Theorem 3

P-value is fraction of perturbations with model's $\overline{P(\text{Success})}$ less than committee's.

- ★ Model's set must be exogenously chosen, e.g. using $\hat{\theta}_S^U$ from a different dataset

Deviation Gains Correction

- Observed deviation gain is combination of true gain and “winner’s curse”

$$\begin{aligned} g(\hat{\theta}_S^U) &= \underbrace{\bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U)}_{\text{Model}} - \underbrace{\bar{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{Committee}} \\ &= \underbrace{\bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U) - \bar{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U)}_{\text{Winner's curse}} + \underbrace{\bar{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U) - \bar{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{True gain at } \hat{\theta}_S^U} \end{aligned}$$

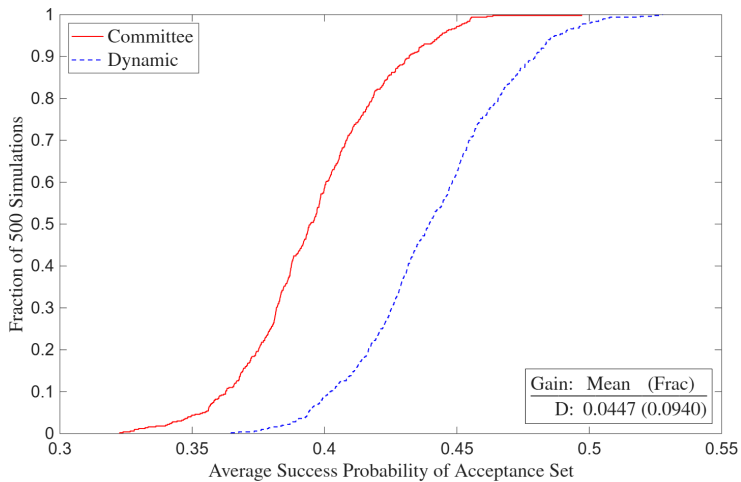
- Subtract estimate of winner’s curse from observed deviation gain

$$\widehat{\text{Winner's curse}} = \frac{1}{B} \sum_{b=1}^B [\bar{P}_{\mathcal{A}^R(\tilde{\theta}_{S,b}^U)}(\tilde{\theta}_{S,b}^U) - \bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\theta}_{S,b}^U)]$$

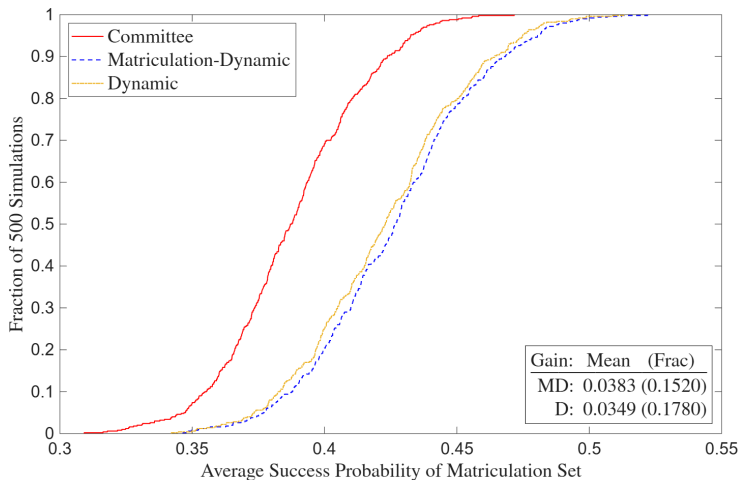
Theorem 4

Correction is exact under certain conditions and removes at least $\sqrt{2} - 1$ of the curse.

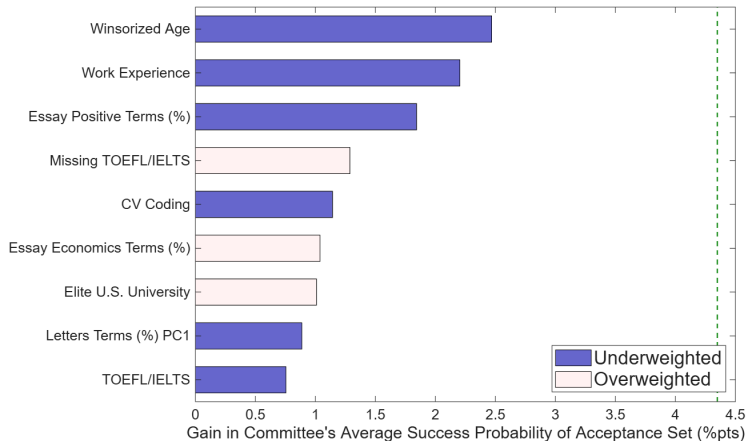
Deviation Gains (Acceptance)



Deviation Gains (Matriculation)



Deviation Gains Attribution (Acceptance)

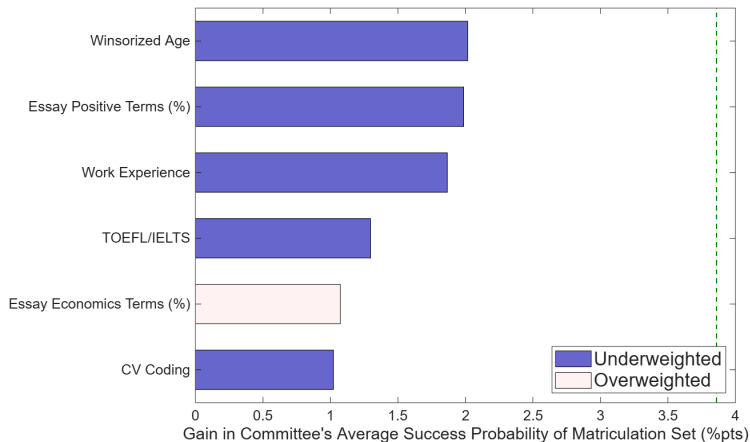


- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

» Full Deviation Gains Attribution (Acceptance)

» Deviation Gains Attribution Table (Acceptance)

Deviation Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

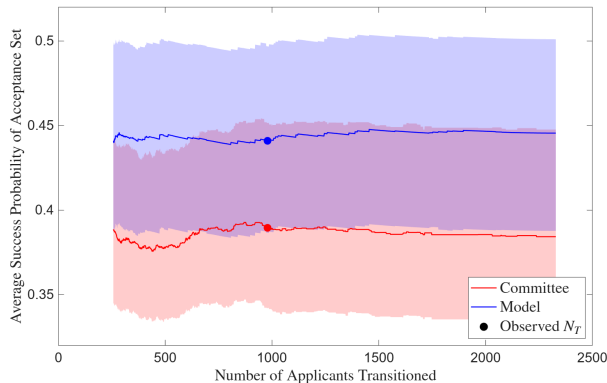
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Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

- Blue = underweighted, white = correctly weighted, red = overweighted
- AMEs don't reveal if committee deviates from optimal ranking near the cutoff

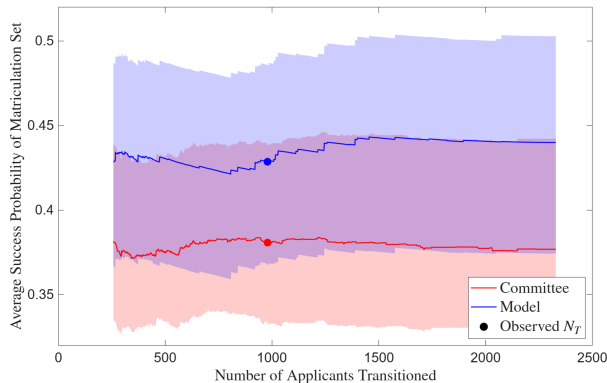
» Acceptance vs. Success AMEs

Benefit of Reading More Files (Acceptance)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_A \bar{P}(n) - cn$
- FOC: $V = c / (N_A \bar{P}'(N_T))$
- V is the dollar value of an additional expected success

Benefit of Reading More Files (Matriculation)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_M \bar{P}(n) - cn$
- FOC: $V = c / (N_M \bar{P}'(N_T))$
- V is the dollar value of an additional expected success

Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to test optimality and quantify deviation gains
 - ① For most applicants, estimated success probabilities align with those implied by committee's decisions. But some applicant characteristics are misweighted
 - ② Correctly weighting these characteristics would increase admitted cohort's average success probability by about 4 percentage points without reading more files
 - ③ Success is hard to predict, but even modest gains in selection are worthwhile
- ★ Methods apply to undergraduate/professional admissions, hiring, and other settings

Thank You!

A1: Synthetic Rows $\neq$ Actual Applicants

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	89	0	0	0	0	5	150
1	0	1	0	1	96	0	1	0	0	6	23
0	0	0	0	0	88	0	1	0	1	.	150
0	0	0	0	0	80	1	1	0	1	.	150
0	0	0	0	1	99	0	1	0	0	.	29
0	0	0	0	0	81	0	0	1	1	.	23
1	0	1	1	1	92	0	1	0	0	3	37
1	1	1	1	1	93	0	0	0	1	3	37
0	0	0	0	0	83	1	0	1	0	.	48
1	1	0	0	0	87	0	1	0	1	5	34



Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	77	0	0	0	1	.	30
0	0	0	0	1	89	0	0	0	0	.	25
0	0	0	0	0	85	1	0	0	0	.	150
0	0	0	0	0	76	0	1	0	1	.	51
1	1	0	0	0	88	0	1	0	0	8	150
1	0	1	0	0	88	0	0	0	1	5	150
1	1	0	0	0	84	0	0	1	0	2	20
0	0	0	0	0	82	0	1	0	0	.	35
0	0	0	0	0	70	1	0	0	0	.	150
1	1	0	0	1	93	0	1	0	0	5	57

A1: Dependent Variables

Table 1a: Dependent Variables

Admission	Success
1(Transition)	1(Complete Program)
<i>1(Accept)</i>	<i>1(Academic Placement)</i>
1(Waitlist)	<i>1(Government Placement)</i>
1(Open House)	<i>1(Consulting Placement)</i>
<i>1(Matriculate)</i>	<i>1(Tech Placement)</i>
1(Attend Program)	<i>1(Other Placement)</i>
Program Rank	<i>1(No Placement)</i>
	Placement Rank
	<i>1(Top 100 Placement)</i>

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
 - For non-tenure track, use weighted average of university's and maximum rankings
 - For non-academia, use average of ChatGPT, Claude, and Gemini's [rankings](#)

A1: Objective Variables

Table 1b: Objective Independent Variables

Demographic	Performance
Year	GRE Verbal Percentile
1 (<i>Covid Year</i>)	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
1 (Female)	Undergraduate GPA
1 (U.S.)	TOEFL/IELTS
1 (<i>U.S. African/Hispanic/Native</i>)	1 (Elite U.S. University)
1 (<i>U.S. Asian</i>)	1 (Elite U.S. Liberal Arts College)
1 (<i>U.S. Other</i>)	1 (Other U.S. University/College)
1 (International Asian)	1 (Elite Foreign University)
1 (International Other)	1 (Other Foreign University)
	1 (Graduate Degree)
	1 (Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	<i>Concentration Dummy Variables</i>

A1: Subjective Variables

Table 1c: Subjective Independent Variables

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	#Math Courses
Letters Length	1(Advanced Math)
Positive Terms (%) (YS 2012)	CV Length
Negative Terms (%) (YS 2012)	CV Coding
Standout Terms (%) (BEMS 2022)	CV 1(Publication/Working Paper)
Ability Terms (%) (BEMS 2022)	Essay Length
Research Terms (%) (BEMS 2022)	Essay 1(Specific Professor)
Grindstone Terms (%) (BEMS 2022)	Essay Economics Terms (%)
Teaching Terms (%) (BEMS 2022)	Essay Research Terms (%)
Communal Terms (%) (BEMS 2022)	Essay Positive Terms (%)
Agentic Terms (%) (MHM 2009)	Essay Negative Terms (%)
	Field Dummy Variables

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

A1: Dictionaries

● Recommendation Letters:

- **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop

● CVs and Essays:

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
 - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
 - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for “grindstone terms” (e.g. depend*, diligen*)
- Research questions emphasize prediction over inference
 - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
 - If GRE predicts success, committee should use it regardless of why

A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
 - Let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$
 - **Step 1:** Run probit of d on z and compute inverse Mills $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
 - **Step 2:** Run OLS or heckprobit of y on $(x, \hat{\lambda})$ to get $\hat{\beta}$
 - **Instruments:** Variables that predict matriculation but not the outcome

A1: Acceptance, Matriculation, and Success AMEs

Dependent Variable	Accept	Matriculate	T100 Placement
Winsorized Age	-0.0001	-0.0223**	0.0053*
Female	0.0163**	-0.0490	-0.0032
U.S.	-0.0013	0.1472*	0.0192
International Asian	-0.0214***	0.1644**	-0.0127
GRE Quantitative Percentile	0.0011**	-0.0147**	0.0006
Undergraduate GPA	0.0195	-0.1026	-0.1096**
TOEFL/IELTS	-0.0022	-0.0631	0.0409*
Work Experience	0.0140**	0.1591**	0.0358**
Committee Score	0.0788***	-0.0371	0.0007
Advanced Math	0.0246	-0.1950**	-0.0287
Letters Terms (%) PC1	0.0055	0.0671**	0.0159
CV Coding	-0.0044*	-0.0414**	0.0125
Essay Economics Terms (%)	0.0001	0.0366**	-0.0104
Essay Positive Terms (%)	-0.0013	-0.0072	0.0123*
Open House		0.2615***	
Program Rank			-0.0011***
Missing Program Rank			-0.3965***

Year Dummies
 GRE Verbal Percentile
 GRE Analytic Percentile
 Elite U.S. University
 Elite U.S. Liberal Arts College
 Elite Foreign University
 Other Foreign University
 Graduate Degree
 #Professor Recommenders
 #Prolific Recommenders
 #Math Courses
 Letters Length/100
 Letters Terms (%) PC2
 CV Length/100
 CV Publication/Working Paper
 Essay Length/100
 Essay Specific Professor
 Essay Research Terms (%)
 Essay Negative Terms (%)
 Missing Dummies

A1: Insufficiency of Comparing AMEs

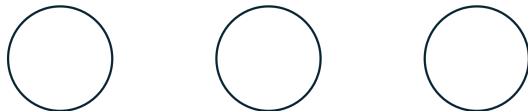
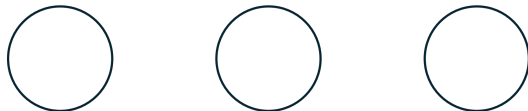
- Cannot test optimality by comparing AMEs across regressions. Suppose:
 - Work experience has large effect on success, graduate degree smaller effect
 - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
 - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
 - **Acceptance Regression:** $AME_{\text{Experience}} = 1, AME_{\text{Degree}} = 0$
 - **Success Regression:** $1 > AME_{\text{Experience}} > AME_{\text{Degree}} > 0$
 - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

A1: Diversity Objective

- **Demographics:** Squares vs. circles

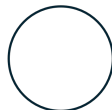
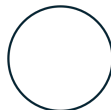
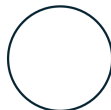
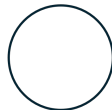


- **Quality:** High vs. low



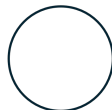
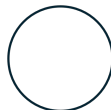
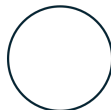
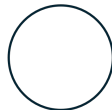
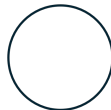
A1: Diversity Objective

- No preference for squares or circles



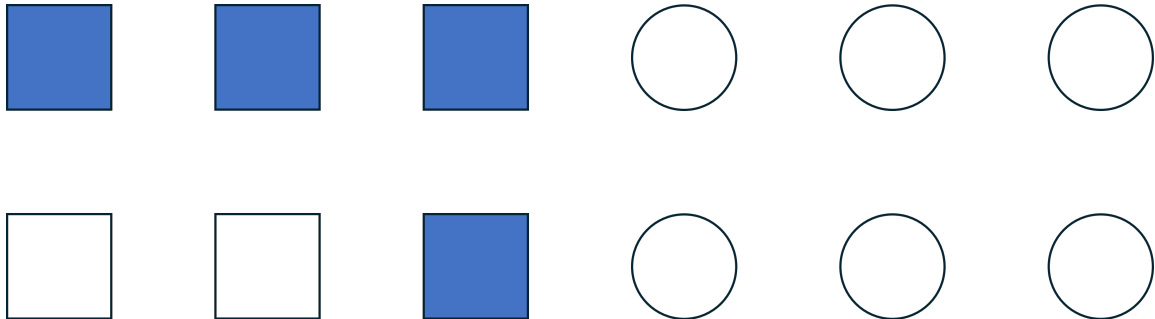
A1: Diversity Objective

- Small preference for squares $\Rightarrow$ consistent with maximizing quality



A1: Diversity Objective

- Large preference for squares $\Rightarrow$ inconsistent with maximizing quality



A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
 - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
 - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
 - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a “success” model
 - **Problem:** Not a test if admissions decisions are optimal
 - Need structural model of admissions to conclude about optimality

A1: Logistic Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$. Let $P(Y_n^E = 1|x_n^E; \theta^E) = \frac{\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}{1 + \exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}$,

and $P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE}) = \frac{\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}{1 + \exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U(y|x; \theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E; \theta^E)^{Y_n^E} [1 - P(Y_n^E = 1|x_n^E; \theta^E)]^{1-Y_n^E} \prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})^{Y_n^{NE}} [1 - P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E} \sum_{n=1}^{N_E} \hat{P}_n^E (1 - \hat{P}_n^E) \hat{\theta}_1^E - \frac{1}{N_{NE}} \sum_{n=1}^{N_{NE}} \hat{P}_n^{NE} (1 - \hat{P}_n^{NE}) \hat{\theta}_1^{NE}$. Then:
 $W = g(\hat{\theta})' [\nabla_{\theta} g(\hat{\theta}) \hat{V}(\hat{\theta}) \nabla_{\theta} g(\hat{\theta})']^{-1} g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

A1: Linear Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$, i.e. (Economics $\perp\!\!\!\perp$ Non-Economics) | Predictors.

Let $f(y_n^E|x_n^E; \beta^E, \sigma_E^2) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E)^2\right]$, and

$f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE})^2\right]$. Then:

$$\mathcal{L}_N^U(y|x; \beta, \sigma^2) = \prod_{n=1}^{N_E} f(y_n^E|x_n^E; \beta^E, \sigma_E^2) \prod_{n=1}^{N_{NE}} f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2).$$

- $Y \in \{\text{Standardized Committee Score, Standardized Placement Rank}\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.)
Then $W = g(\hat{\beta}_1)'[\nabla_{\beta, \sigma^2} g(\hat{\beta}_1) \hat{V}(\hat{\beta}, \hat{\sigma}^2) \nabla_{\beta, \sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n, \sigma_n^2 = \sigma^2$)

A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
 - Need different skills at different stages in the program, such as grades vs. placement
 - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
 - **Admission:** May have different preparation and/or interests
 - **Completion:** Depending on country, may have fewer outside options
 - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
 - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
 - For economics, increased importance of elite predocs and Fed RAs

A2: Models of Admissions

- ① **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
 - ① **Static Model:** In Round 1, rank applicants based only on objective variables
 - ② **Myopic Model:** Like dynamic, but distribution of z not conditional on x
- ② **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)

A2: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z, t, \bar{k})$, where Success = $\mathbb{1}(\text{Complete Program})$ or $\mathbb{1}(\text{Good Placement})$. If reject applicant, get $P(y_2^* = S|t) = P(\text{Success})$ of year t 's marginal accepted applicant
- **Round 1:** If reject applicant, get $L/P(y_1^* = S|t) = L/P(\text{Success})$ of year t 's marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort
 - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] =$ expected value of Round 2, where z is integrated out conditional on (x, t) based on $F(z|x, t)$
 - **Static:** If transition applicant, get $P(y = S|x, t, \bar{k})$, or $P(\text{Success})$ unconditional on z . It may differ from $P(y = S|x, z, t, \bar{k})$, though it still accounts for z 's effect on y
 - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$, replace $F(z|x, t)$ with $F(z|t)$

A2: Portfolio Choice to Cutoff Rule (BEMS 2022)

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda^* \end{cases}$$

Upshot: λ^* is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2.1: Portfolio Choice to Cutoff Rule with Shocks

Portfolio Choice Problem:

$$\begin{aligned} \max_{(d_1, \dots, d_N) \in \{A, R\}^N} & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + \epsilon_n(R) \mathbb{1}(d_n = R) \}, \\ \text{such that: } & \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \dots, d_N; \lambda) = \\ & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + [\lambda + \epsilon_n(R)] \mathbb{1}(d_n = R) \} + \lambda(N_A - N) \end{aligned}$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n, \epsilon_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

Theorem 5

For large N , there is a cutoff rule solution with logistic CCPs.

► Portfolio Choice Problem

A2: Portfolio Choice to Cutoff Rule, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] + \lambda_2 (N_A - N_T)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(x_n, z_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda_2^* \end{cases}$$

Upshot: λ_2^* is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2: Portfolio Choice to Cutoff Rule, Round 1 (Dynamic)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n), \bar{\lambda}_2^*\} dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff level. Denote it hereafter as $L(y_1^* = S) \in [EV_{2,N_T+1}, EV_{2,N_T}]$

A2: Portfolio Choice to Cutoff Rule, Round 1 (Static)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S | x_n) > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S | x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S | x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in [P(y_{N_T+1} = S | x_{N_T+1}), P(y_{N_T} = S | x_{N_T})]$

A2: Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, t; \theta_A) = \frac{\exp(f_A(x, z, t)\theta_A)}{1 + \exp(f_A(x, z, t)\theta_A)}$$

Success Probability:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P(y_2^* = S | t; \theta_{S_2^*}) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

A2: Dynamic Parameterization 2

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \tilde{d}_2)/\sigma} \right) dF(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t)\end{aligned}$$

- **Continuous:** $f_{z|x,t}(z_j|x, t; \beta_{F_{z|x,t,j}}, \sigma_{F_{z|x,t,j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x,t,j}}^2}} \exp \left[\frac{-1}{2\sigma_{F_{z|x,t,j}}^2} (z_j - g(x, t)\beta_{F_{z|x,t,j}})^2 \right]$
- **Binary:** $P(z_j = 1|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}})}{1 + \exp(g(x, t)\theta_{F_{z|x,t,j}})}$
- **Multinomial:** $P(z_j = \ell|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}}^\ell)}{1 + \sum_{i=1}^{L_j-1} \exp(g(x, t)\theta_{F_{z|x,t,j}}^i)}, \ell = 1, \dots, L_j - 1$

A2: Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\
 & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Success mostly unobservable for applicants from fewer than six years ago

A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A) \mathbb{1}(d_{2,n}=A)}_{\text{Conditional choice probabilities}} \\
 & \underbrace{(1 - P_{d_{2,n}}(\theta_A)) \mathbb{1}(d_{2,n}=R)}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^N \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{f\left(z_n^{-c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{-c}|x, t} \\
 & \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{P_{y_n}(\theta_S) \mathbb{1}(y_n=S) (1 - P_{y_n}(\theta_S)) \mathbb{1}(y_n=F)}_{\text{Success probabilities}}
 \end{aligned}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

- **Motivation:** Interpretable cutoff probabilities in both rounds

» Dynamic Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

» Dynamic Model

Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S | x, t, \bar{k}) \mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S|x, t, \bar{k})\mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- $P(y = S|x, t, \bar{k})$ estimates inconsistent, but fitted values account for z 's effect on y

A2: Static Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \\ \prod_{n=1}^N \underbrace{[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})]^{\mathbb{1}(y_n=S)} [(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success mostly unobservable for applicants from fewer than six years ago

A2: Cutoff Rule with Matriculation

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^N P(m_n = M | w_n) \mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | w_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] P(m_n = M | w_n) + \lambda(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(w_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda^* \\ R & \text{if } P(y_n = S | w_n) < \lambda^* \end{cases}$$

A2: Cutoff Rule with Matriculation (Large N)

Portfolio Choice Problem:

$$\max_{\delta \in \{\delta: \mathbb{R}^K \rightarrow [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w)P(m = M|w)\delta(w)f(w)dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w)\delta(w)f(w)dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} [P(y = S|w)\delta(w) + \lambda(1 - \delta(w))]P(m = M|w)f(w)dw + \lambda(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K, \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0, 1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

Upshot: Point-identified cutoff probability λ^* is with respect to success probabilities

A2: Cutoff Rule with Matriculation, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | w_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] P(m_n = M | w_n) + \lambda_2 (N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(w_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | w_n) < \lambda_2^* \end{cases}$$

A2: Cutoff Rule with Matriculation, Round 1

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n) - \bar{\lambda}_2^*, 0\} P(m_n = M | x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

A2: Matriculation-Dynamic Likelihood

Matriculation Probability:

$$P_m = P(m = M | x, z, t; \theta_M) = \frac{\exp(f_M(x, z, t)\theta_M)}{1 + \exp(f_M(x, z, t)\theta_M)}$$

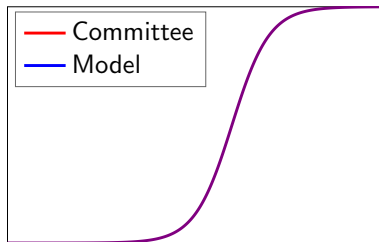
$$\begin{aligned} \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_A} \underbrace{P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)} (1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)}}_{\text{Matriculation probabilities}} \\ & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\ & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}} \end{aligned}$$

A2: Empirical Considerations

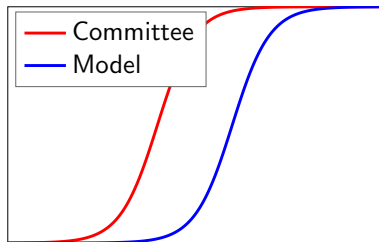
- **Selection Effects:** Track success for all applicants, including non-matriculants
 - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
 - In decision blocks containing success, fix *Program Rank* at this university's rank
 - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
 - **Example:** Recommendation letter has nuances that algorithms can't detect
 - Include subjective committee score in z as a proxy for this information

A2: How Deviation Gains Test Works

- 1 Write Bellman equations for committee's optimization problem
- 2 Estimate parameters. Rationality assumption distorts estimates
- 3 Re-estimate parameters without rationality, and use to solve model
- 4 Compare selections from model's decision rule to actual decision rule
- 5 To prevent overfitting, use perturbations of estimates for comparison



Optimal Committee



Suboptimal Committee

A2: Deviation Gains Algorithm

Dynamic/Static/Myopic:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic and myopic, or $P(y = S|x, t, \bar{k})$ for static. Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$, and accept top N_A survivors.
Compute $P(y = S|x, z, t, \bar{k})$ of acceptance set, and compare to committee's set

Matriculation-Dynamic/Static:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \bar{k})$ for static (best analogue). Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$ and accept them in descending order, each matriculating with probability $P(m = M|x, z, t)$, until N_M matriculate.
Compute $P(y = S|x, z, t, \bar{k})$ of matriculation set, and compare to committee's set

A2: Randomization Test

Theorem 6

Let d^R be the model's decision rule, and d^U the actual decision rule. By optimality:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1} \{ d_{1,n} [P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^R [P_{y_n}(\hat{\theta}_S^U)] = A \} \\ \geq \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1} \{ d_{1,n} [P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^U [P_{y_n}(\hat{\theta}_A^U)] = A \} \end{aligned}$$

- But for perturbation $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$'s asymptotic distribution, can have:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \tilde{\theta}_{S,b}^U) \mathbb{1} \{ d_{1,n} [P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^R [P_{y_n}(\hat{\theta}_S^U)] = A \} \\ < \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \tilde{\theta}_{S,b}^U) \mathbb{1} \{ d_{1,n} [P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^U [P_{y_n}(\hat{\theta}_A^U)] = A \} \end{aligned}$$

- For matriculation models, compare matriculants, not acceptances

A2: Deviation Gains Interpretation

- Under meaningfully-sized deviation gains, committee may have incorrect beliefs about which types of applicants are most likely to succeed
 - Too few observations for a true out-of-sample test, and cannot account for changes in the determinants of success over time, such as writing essays with AI
- Committee may use success-predicting variables that I do not observe
 - Because the success probability measure used to evaluate gains cannot detect such variables, any edge the committee's private information provides goes unmeasured
 - Comprehensive data, especially committee score in z , protects against this limitation

A2: Summary of Simulated Results

- **Simulated Results:** Estimation results are well behaved
 - **Dynamic/Static:** Deviation gains larger for simulated dataset where committee misweights applicant characteristics than for optimal simulated dataset
 - **Matriculation-Dynamic/Static:** Same as above
- Example of cutoff probability reversal if committee transitions few files in Round 1:
 - Three applicants: A, B, and C. Committee transitions two, accepts one
 - A is best and C is worst overall, but A is worst in objective variables
 - Committee rejects A in Round 1 and accepts B, who is worse than A

A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
 - **Example:** Applicants with work experience more likely to succeed and matriculate
 - Can help committee find “people that we can afford” (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
 - Instead maximize combination of $P(\text{Success})$ and other objectives
 - **Example:** Committee may want diverse cohort (e.g. across fields):
 $P(\text{Success})^\alpha \times \text{Diversity}^{1-\alpha}$, where $\alpha \in [0, 1]$ can be estimated
- **Extension 3:** Waitlist and multiyear models (see [online appendix](#))
 - Three-round models, and two-round models that repeat over infinite horizon
 - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

A3: Economics Department Summary Statistics

Table 2: Economics Department Summary Statistics

(a) Objective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1820	0.3973	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Winsorized Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6387	0.3026	3.6645	0.3063	3.7227	0.2610	3.7068	0.2880
TOEFL IELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Profic Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL/IELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Winsorized Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	2329		979		257		79	

(b) Subjective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2031
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

A3: Economics Department Correlation Matrix

Table 3: Economics Department Correlation Matrix

[illegible]

A3: Economics Department Rec Letters Principal Components

Table 4: Economics Department Rec Letters Principal Components

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

A3: Economics Department Logistic Regression Results

Table 5a: Economics Department Logistic Regression Results

[illegible]

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A3: Economics Department Logistic Regression AMEs

Table 5b: Economics Department Logistic Regression AMEs

Explanatory Variable	Intercept	Female	International Asian	GRE Quantitative Percentile	GRE Analytical Percentile	Work Experience	Committee Score	CV Coding	CV Publishing/Working Paper	Essay Positive Terms (%)	Program Rank	Open House
Intercept	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Female		0.000										
International Asian			0.000									
GRE Quantitative Percentile				0.000								
GRE Analytical Percentile					0.000							
Work Experience						0.000						
Committee Score							0.000					
CV Coding								0.000				
CV Publishing/Working Paper									0.000			
Essay Positive Terms (%)										0.000		
Program Rank											0.000	
Open House												0.000

- **Accept** ($p < .1$): 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success** ($p < .1$): Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A3: Economics Department Logistic Regression Results (Synthetic)

Table 6a: Economics Department Logistic Regression Results (Synthetic)

Explanatory Variable	Intercept	Age	Age ²	Age ³	Female	Elite U.S. University	Missing TOEFL/IELTS	Program Rank	Program Rank ²
Intercept	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Age	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Age ²	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Age ³	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Female	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Elite U.S. University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Missing TOEFL/IELTS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Program Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Program Rank ²	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Intercept	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Age	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Age ²	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Age ³	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Female	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Elite U.S. University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Missing TOEFL/IELTS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Program Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Program Rank ²	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

- **Accept** ($p < .1$): 2017 (-), Winsorized Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success** ($p < .1$): Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A3: Economics Department Logistic Regression AMEs (Synthetic)

Table 6b: Economics Department Logistic Regression AMEs (Synthetic)

Explanatory Variable	Intercept	Gender	Age	Letters	PC2	Missing	Committee	Score	TOEFL	IELTS	Math	Coding	Program	Rank
Intercept	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Gender	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Age	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Letters	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
PC2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Missing	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Committee	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Score	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
TOEFL	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
IELTS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Math	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Coding	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Program	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

- **Accept** ($p < .1$): 2017 (-), Winsorized Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success** ($p < .1$): Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A3: Economics Department Lasso Logistic Regression Results

Table 7a: Economics Department Lasso Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1107								
2017		-0.4928							
2018		-2.4456	0.0186						
2019		-1.7752							
Winsorized Age				0.0084	0.0047	0.0184			
Female	0.2674	0.5589							
U.S.	-0.1959						0.9477		
International Asian	-0.8666	-0.8651	0.1271				-0.5295	-0.7927	
GRE Verbal Percentile	0.0227	0.0047	-0.0015	0.0015			-0.0085		
GRE Quantitative Percentile	0.1027	0.0174	-0.0273	0.0179	0.0125		-0.2706	-0.2243	-0.0070
GRE Analytic Percentile	0.0056	0.0035							
Undergraduate GPA	0.9612	0.7135				-0.1316			0.2561
TOEFL/IELTS	0.3055		-0.2943	0.0300		0.0934	-0.6903	-0.5850	-2.2892
Elite U.S. University	0.4024								
Elite U.S. Liberal Arts College	0.1112								
Elite Foreign University	-0.4025								
Other Foreign University	0.0790	0.0940	0.1998						-0.2265
Missing Undergraduate Institution			-0.0036						
Graduate Degree	0.1178								
Work Experience	0.6006	0.4068	0.0404	0.0961	0.0470	0.1976			-2.0809
#Professor Recommenders									0.0022
#Prolific Recommenders	0.1534	0.0301	-0.0870	0.0526	0.0363		-0.0789	-0.0358	
Missing GRE	-0.0534			-0.0697					
Missing Undergraduate GPA	-0.2348	-0.1148							
Missing TOEFL/IELTS	0.1447			-0.1243		-0.1295	1.8402	2.2548	
Committee Score		3.4243				0.0353		-0.4899	
#Math Courses	-0.0553	-0.0354							
Advanced Math	0.4302	0.7352							
Letters Length/100	0.1196	0.0587							
Letters Terms (%) PC1	0.0746	0.1485	0.1258			0.0102			
Letters Terms (%) PC2	-0.0162	-0.2001							
CV Length/100	0.0219	0.0298		0.0014		-0.0286	-0.1179		
CV Coding	-0.0038	-0.1754		0.0165	0.0076	-0.2540	-0.4154	-0.0998	
CV Publication/Working Paper	0.5563	0.6436		0.0892		-0.0987			
Essay Length/100	0.1202	0.0295							
Essay Specific Professor	0.3403	0.3958	0.0674						
Essay Economics Terms (%)	-0.0016			-0.0405	-0.0325	-0.0104	0.4473	0.6570	0.2117
Essay Research Terms (%)		-0.0278						-0.1506	
Essay Positive Terms (%)	-0.0049	-0.0382		0.0192	0.0140	0.0267			
Essay Negative Terms (%)		-0.0417							
Missing Committee Score		-0.9416			-0.2615	-0.0714	4.0146	0.7971	
Missing Application Pdf	-1.0365								
Open House			1.0982						
Program Rank						-0.0082			0.1385
Missing Program Rank						-2.2557			19.4273
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2320	2320	257	2320	2320	2320	2320	2320	2320
Pseudo R ²	0.2841	0.3807	0.1281	0.0216	0.0265	0.0276	0.0276	0.2693	
CV Pseudo R ²	0.2441	0.7605	0.0596	0.0082	0.0129	0.0199	0.0121	0.0183	0.2638

- Accept:** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Verbal Percentile (+), GRE Quantitative Percentile (+), GRE Analytic Percentile (+), Undergraduate GPA (+), Elite U.S. Liberal Arts College (+), Elite Foreign University (-), Other Foreign University (+), Work Experience (+), # Prolific Recommenders (+), Missing Undergraduate GPA (-), Committee Score (+), #Math Courses (-), Advanced Math (+), Letters Length/100 (+), Letters Terms (%) PC1 (+), Letters Terms (%) PC2 (-), CV Length/100 (+), CV Coding (-), CV Publication/Working Paper (+), Essay Length/100 (+), Essay Specific Professor (+), Economics Research Terms (%) (-), Essay Positive Terms (%) (-), Essay Negative Terms (%) (-), Missing Committee Score (-)
- Matriculate:** 2018 (+), International Asian (+), GRE Verbal Percentile (-), GRE Quantitative Percentile (-), TOEFL/IELTS (-), Other Foreign University (+), Missing Undergraduate Institution (-), Work Experience (+), #Prolific Recommenders (-), Letters Terms (%) PC1 (+), Essay Specific Professor (+), Open House (+)
- T100 Placement:** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Letters Terms (%) PC1 (+), Essay Economics Terms (%) (-), Essay Positive Terms (%) (+), Missing Committee Score (-), Program Rank (-), Missing Program Rank (-)

A3: Economics Department Dynamic Estimates

Table 8: Economics Department Dynamic Estimates

	Conjunction Tests		T-Statistic		Asymptotic		Sum of Squares	
	Chi-Square	df	Statistic	df	Chi-Square	df	Statistic	df
2005	2.062**	1	0.000	1	0.000	1	0.000	1
2006	2.070**	1	0.048	1	0.001	1	0.013	1
2007	2.066	1	0.006	1	0.000	1	0.000	1
2008	0.000	1	0.000	1	0.000	1	0.000	1
2017	0.000	1	0.003	1	0.000	1	0.000	1
2018	0.000	1	0.000	1	0.000	1	0.000	1
2019	0.000	1	0.000	1	0.000	1	0.000	1
2020	0.000	1	0.000	1	0.000	1	0.000	1
2021	0.000	1	0.000	1	0.000	1	0.000	1
2022	0.000	1	0.000	1	0.000	1	0.000	1
Mean Age	0.000**	1	0.000	1	0.000	1	0.000	1
Female	0.000**	1	0.000	1	0.000	1	0.000	1
U.S.	0.000**	1	0.000	1	0.000	1	0.000	1
International	0.000**	1	0.000	1	0.000	1	0.000	1
Work Potential	0.000**	1	0.000	1	0.000	1	0.000	1
Q2: Virtual Presence	0.000**	1	0.000	1	0.000	1	0.000	1
Q3: Quality Perceptions	0.000**	1	0.000	1	0.000	1	0.000	1
Q4: Affective Perceptions	0.000**	1	0.000	1	0.000	1	0.000	1
TDSPI: ITC	0.000**	1	0.000	1	0.000	1	0.000	1
TDSPI: ITC2	0.000**	1	0.000	1	0.000	1	0.000	1
U.S. 113 School Age	0.000**	1	0.000	1	0.000	1	0.000	1
U.S. Foreign College	0.000**	1	0.000	1	0.000	1	0.000	1
Other Foreign College	0.000**	1	0.000	1	0.000	1	0.000	1
Other Foreign Institution	0.000**	1	0.000	1	0.000	1	0.000	1
College Degree	0.000**	1	0.000	1	0.000	1	0.000	1
Work Experience	0.000**	1	0.000	1	0.000	1	0.000	1
Public Recommendations	0.000**	1	0.000	1	0.000	1	0.000	1
Public Recommendations	0.000**	1	0.000	1	0.000	1	0.000	1
Using QR	0.000**	1	0.000	1	0.000	1	0.000	1
Using TDSPI, ITC	0.000**	1	0.000	1	0.000	1	0.000	1
Conjunction Tests	0.000**	1	0.000	1	0.000	1	0.000	1
Math Counts	0.000**	1	0.000	1	0.000	1	0.000	1
Advanced Math	0.000**	1	0.000	1	0.000	1	0.000	1
Letters Length: LO	0.000**	1	0.000	1	0.000	1	0.000	1
Letters Length: FC1	0.000**	1	0.000	1	0.000	1	0.000	1
Letters Length: FC2	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: LO	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC1	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC2	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC3	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC4	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC5	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC6	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC7	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC8	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC9	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC10	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC11	0.000**	1	0.000	1	0.000	1	0.000	1
CV Length: FC12	0.000**	1	0.000	1	0.000	1	0.000	1

[illegible]

Obs	Form	LL	AE	2020	802	2020.8	2020.1
Subject standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.							

Obs Param (LL ΔC	2020	2021	2022	2023
ΔC (mm) <td>0.00</td> <td>0.00</td> <td>0.00</td> <td>0.00</td>	0.00	0.00	0.00	0.00

A3: Restricted Static Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	29.36% (23.20%, 36.38%)	39.20% (32.75%, 46.05%)	
⋮	⋮	⋮	
2019	34.04% (28.67%, 39.85%)	39.62% (32.54%, 47.16%)	
Sigma			0.0064 (0.0035)

- Static model has interpretable Round 1 cutoff probabilities
- Round 2 cutoff probabilities always greater than those in Round 1

A3: Economics Department Static Estimates

Table 9: Economics Department Static Estimates

[illegible]

►► Restricted Dynamic Estimates

A3: Economics Department Matriculation-Dynamic Estimates

Table 11: Economics Department Matriculation-Dynamic Estimates

(a) Unrestricted

(b) Restricted

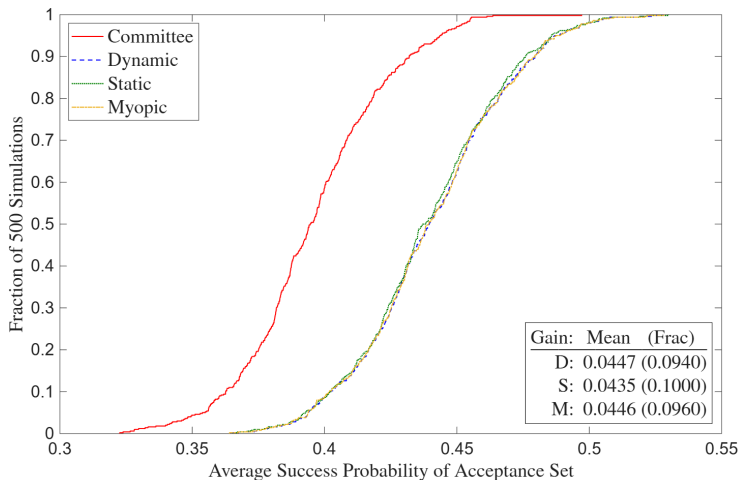
[illegible]

	Conf 1	Conf 2	Conf 3	Conf 4	Conf 5	Conf 6	Conf 7	Conf 8	Conf 9	Conf 10	Conf 11	Conf 12	Conf 13	Conf 14	Conf 15	Conf 16	Conf 17	Conf 18	Conf 19	Conf 20	Conf 21	Conf 22	Conf 23	Conf 24	Conf 25	Conf 26	Conf 27	Conf 28	Conf 29	Conf 30	Conf 31	Conf 32	Conf 33	Conf 34	Conf 35	Conf 36	Conf 37	Conf 38	Conf 39	Conf 40	Conf 41	Conf 42	Conf 43	Conf 44	Conf 45	Conf 46	Conf 47	Conf 48	Conf 49	Conf 50	Conf 51	Conf 52	Conf 53	Conf 54	Conf 55	Conf 56	Conf 57	Conf 58	Conf 59	Conf 60	Conf 61	Conf 62	Conf 63	Conf 64	Conf 65	Conf 66	Conf 67	Conf 68	Conf 69	Conf 70	Conf 71	Conf 72	Conf 73	Conf 74	Conf 75	Conf 76	Conf 77	Conf 78	Conf 79	Conf 80	Conf 81	Conf 82	Conf 83	Conf 84	Conf 85	Conf 86	Conf 87	Conf 88	Conf 89	Conf 90	Conf 91	Conf 92	Conf 93	Conf 94	Conf 95	Conf 96	Conf 97	Conf 98	Conf 99	Conf 100	Conf 101	Conf 102	Conf 103	Conf 104	Conf 105	Conf 106	Conf 107	Conf 108	Conf 109	Conf 110	Conf 111	Conf 112	Conf 113	Conf 114	Conf 115	Conf 116	Conf 117	Conf 118	Conf 119	Conf 120	Conf 121	Conf 122	Conf 123	Conf 124	Conf 125	Conf 126	Conf 127	Conf 128	Conf 129	Conf 130	Conf 131	Conf 132	Conf 133	Conf 134	Conf 135	Conf 136	Conf 137	Conf 138	Conf 139	Conf 140	Conf 141	Conf 142	Conf 143	Conf 144	Conf 145	Conf 146	Conf 147	Conf 148	Conf 149	Conf 150	Conf 151	Conf 152	Conf 153	Conf 154	Conf 155	Conf 156	Conf 157	Conf 158	Conf 159	Conf 160	Conf 161	Conf 162	Conf 163	Conf 164	Conf 165	Conf 166	Conf 167	Conf 168	Conf 169	Conf 170	Conf 171	Conf 172	Conf 173	Conf 174	Conf 175	Conf 176	Conf 177	Conf 178	Conf 179	Conf 180	Conf 181	Conf 182	Conf 183	Conf 184	Conf 185	Conf 186	Conf 187	Conf 188	Conf 189	Conf 190	Conf 191	Conf 192	Conf 193	Conf 194	Conf 195	Conf 196	Conf 197	Conf 198	Conf 199	Conf 200	Conf 201	Conf 202	Conf 203	Conf 204	Conf 205	Conf 206	Conf 207	Conf 208	Conf 209	Conf 210	Conf 211	Conf 212	Conf 213	Conf 214	Conf 215	Conf 216	Conf 217	Conf 218	Conf 219	Conf 220	Conf 221	Conf 222	Conf 223	Conf 224	Conf 225	Conf 226	Conf 227	Conf 228	Conf 229	Conf 230	Conf 231	Conf 232	Conf 233	Conf 234	Conf 235	Conf 236	Conf 237	Conf 238	Conf 239	Conf 240	Conf 241	Conf 242	Conf 243	Conf 244	Conf 245	Conf 246	Conf 247	Conf 248	Conf 249	Conf 250	Conf 251	Conf 252	Conf 253	Conf 254	Conf 255	Conf 256	Conf 257	Conf 258	Conf 259	Conf 260	Conf 261	Conf 262	Conf 263	Conf 264	Conf 265	Conf 266	Conf 267	Conf 268	Conf 269	Conf 270	Conf 271	Conf 272	Conf 273	Conf 274	Conf 275	Conf 276	Conf 277	Conf 278	Conf 279	Conf 280	Conf 281	Conf 282	Conf 283	Conf 284	Conf 285	Conf 286	Conf 287	Conf 288	Conf 289	Conf 290	Conf 291	Conf 292	Conf 293	Conf 294	Conf 295	Conf 296	Conf 297	Conf 298	Conf 299	Conf 300	Conf 301	Conf 302	Conf 303	Conf 304	Conf 305	Conf 306	Conf 307	Conf 308	Conf 309	Conf 310	Conf 311	Conf 312	Conf 313	Conf 314	Conf 315	Conf 316	Conf 317	Conf 318	Conf 319	Conf 320	Conf 321	Conf 322	Conf 323	Conf 324	Conf 325	Conf 326	Conf 327	Conf 328	Conf 329	Conf 330	Conf 331	Conf 332	Conf 333	Conf 334	Conf 335	Conf 336	Conf 337	Conf 338	Conf 339	Conf 340	Conf 341	Conf 342	Conf 343	Conf 344	Conf 345	Conf 346	Conf 347	Conf 348	Conf 349	Conf 350	Conf 351	Conf 352	Conf 353	Conf 354	Conf 355	Conf 356	Conf 357	Conf 358	Conf 359	Conf 360	Conf 361	Conf 362	Conf 363	Conf 364	Conf 365	Conf 366	Conf 367	Conf 368	Conf 369	Conf 370	Conf 371	Conf 372	Conf 373	Conf 374	Conf 375	Conf 376	Conf 377	Conf 378	Conf 379	Conf 380	Conf 381	Conf 382	Conf 383	Conf 384	Conf 385	Conf 386	Conf 387	Conf 388	Conf 389	Conf 390	Conf 391	Conf 392	Conf 393	Conf 394	Conf 395	Conf 396	Conf 397	Conf 398	Conf 399	Conf 400	Conf 401	Conf 402	Conf 403	Conf 404	Conf 405	Conf 406	Conf 407	Conf 408	Conf 409	Conf 410	Conf 411	Conf 412	Conf 413	Conf 414	Conf 415	Conf 416	Conf 417	Conf 418	Conf 419	Conf 420	Conf 421	Conf 422	Conf 423	Conf 424	Conf 425	Conf 426	Conf 427	Conf 428	Conf 429	Conf 430	Conf 431	Conf 432	Conf 433	Conf 434	Conf 435	Conf 436	Conf 437	Conf 438	Conf 439	Conf 440	Conf 441	Conf 442	Conf 443	Conf 444	Conf 445	Conf 446	Conf 447	Conf 448	Conf 449	Conf 450	Conf 451	Conf 452	Conf 453	Conf 454	Conf 455	Conf 456	Conf 457	Conf 458	Conf 459	Conf 460	Conf 461	Conf 462	Conf 463	Conf 464	Conf 465	Conf 466	Conf 467	Conf 468	Conf 469	Conf 470	Conf 471	Conf 472	Conf 473	Conf 474	Conf 475	Conf 476	Conf 477	Conf 478	Conf 479	Conf 480	Conf 481	Conf 482	Conf 483	Conf 484	Conf 485	Conf 486	Conf 487	Conf 488	Conf 489	Conf 490	Conf 491	Conf 492	Conf 493	Conf 494	Conf 495	Conf 496	Conf 497	Conf 498	Conf 499	Conf 500	Conf 501	Conf 502	Conf 503	Conf 504	Conf 505	Conf 506	Conf 507	Conf 508	Conf 509	Conf 510	Conf 511	Conf 512	Conf 513	Conf 514	Conf 515	Conf 516	Conf 517	Conf 518	Conf 519	Conf 520	Conf 521	Conf 522	Conf 523	Conf 524	Conf 525	Conf 526	Conf 527	Conf 528	Conf 529	Conf 530	Conf 531	Conf 532	Conf 533	Conf 534	Conf 535	Conf 536	Conf 537	Conf 538	Conf 539	Conf 540	Conf 541	Conf 542	Conf 543	Conf 544	Conf 545	Conf 546	Conf 547	Conf 548	Conf 549	Conf 550	Conf 551	Conf 552	Conf 553	Conf 554	Conf 555	Conf 556	Conf 557	Conf 558	Conf 559	Conf 560	Conf 561	Conf 562	Conf 563	Conf 564	Conf 565	Conf 566	Conf 567	Conf 568	Conf 569	Conf 570	Conf 571	Conf 572	Conf 573	Conf 574	Conf 575	Conf 576	Conf 577	Conf 578	Conf 579	Conf 580	Conf 581	Conf 582	Conf 583	Conf 584	Conf 585	Conf 586	Conf 587	Conf 588	Conf 589	Conf 590	Conf 591	Conf 592	Conf 593	Conf 594	Conf 595	Conf 596	Conf 597	Conf 598	Conf 599	Conf 600	Conf 601	Conf 602	Conf 603	Conf 604	Conf 605	Conf 606	Conf 607	Conf 608	Conf 609	Conf 610	Conf 611	Conf 612	Conf 613	Conf 614	Conf 615	Conf 616	Conf 617	Conf 618	Conf 619	Conf 620	Conf 621	Conf 622	Conf 623	Conf 624	Conf 625	Conf 626	Conf 627	Conf 628	Conf 629	Conf 630	Conf 631	Conf 632	Conf 633	Conf 634	Conf 635	Conf 636	Conf 637	Conf 638	Conf 639	Conf 640	Conf 641	Conf 642	Conf 643	Conf 644	Conf 645	Conf 646	Conf 647	Conf 648	Conf 649	Conf 650	Conf 651	Conf 652	Conf 653	Conf 654	Conf 655	Conf 656	Conf 657	Conf 658	Conf 659	Conf 660	Conf 661	Conf 662	Conf 663	Conf 664	Conf 665	Conf 666	Conf 667	Conf 668	Conf 669	Conf 670	Conf 671	Conf 672	Conf 673	Conf 674	Conf 675	Conf 676	Conf 677	Conf 678	Conf 679	Conf 680	Conf 681	Conf 682	Conf 683	Conf 684	Conf 685	Conf 686	Conf 687	Conf 688	Conf 689	Conf 690	Conf 691	Conf 692	Conf 693	Conf 694	Conf 695	Conf 696	Conf 697	Conf 698	Conf 699	Conf 700	Conf 701	Conf 702	Conf 703	Conf 704	Conf 705	Conf 706	Conf 707	Conf 708	Conf 709	Conf 710	Conf 711	Conf 712	Conf 713	Conf 714	Conf 715	Conf 716	Conf 717	Conf 718	Conf 719	Conf 720	Conf 721	Conf 722	Conf 723	Conf 724	Conf 725	Conf 726	Conf 727	Conf 728	Conf 729	Conf 730	Conf 731	Conf 732	Conf 733	Conf 734	Conf 735	Conf 736	Conf 737	Conf 738	Conf 739	Conf 740	Conf 741	Conf 742	Conf 743	Conf 744	Conf 745	Conf 746	Conf 747	Conf 748	Conf 749	Conf 750	Conf 751	Conf 752	Conf 753	Conf 754	Conf 755	Conf 756	Conf 757	Conf 758	Conf 759	Conf 760	Conf 761	Conf 762	Conf 763	Conf 764	Conf 765	Conf 766	Conf 767	Conf 768	Conf 769	Conf 770	Conf 771	Conf 772	Conf 773	Conf 774	Conf 775	Conf 776	Conf 777	Conf 778	Conf 779	Conf 780	Conf 781	Conf 782	Conf 783	Conf 784	Conf 785	Conf 786	Conf 787	Conf 788	Conf 789	Conf 790	Conf 791	Conf 792	Conf 793	Conf 794	Conf 795	Conf 796	Conf 797	Conf 798	Conf 799	Conf 800	Conf 801	Conf 802	Conf 803	Conf 804	Conf 805	Conf 806	Conf 807	Conf 808	Conf 809	Conf 810	Conf 811	Conf 812	Conf 813	Conf 814	Conf 815	Conf 816	Conf 817	Conf 818	Conf 819	Conf 820	Conf 821	Conf 822	Conf 823	Conf 824	Conf 825	Conf 826	Conf 827	Conf 828	Conf 829	Conf 830	Conf 831	Conf 832	Conf 833	Conf 834	Conf 835	Conf 836	Conf 837	Conf 838	Conf 839	Conf 840	Conf 841	Conf 842	Conf 843	Conf 844	Conf 845	Conf 846	Conf 847	Conf 848	Conf 849	Conf 850	Conf 851	Conf 852	Conf 853	Conf 854	Conf 855	Conf 856	Conf 857	Conf 858	Conf 859	Conf 860	Conf 861	Conf 862	Conf 863	Conf 864	Conf 865	Conf 866	Conf 867	Conf 868	Conf 869	Conf 870	Conf 871	Conf 872	Conf 873	Conf 874	Conf 875	Conf 876	Conf 877	Conf 878	Conf 879	Conf 880	Conf 881	Conf 882	Conf 883	Conf 884	Conf 885	Conf 886	Conf 887	Conf 888	Conf 889	Conf 890	Conf 891	Conf 892	Conf 893	Conf 894	Conf 895	Conf 896	Conf 897	Conf 898	Conf 899	Conf 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Stamps	4.8E+17	4.8E+17	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							

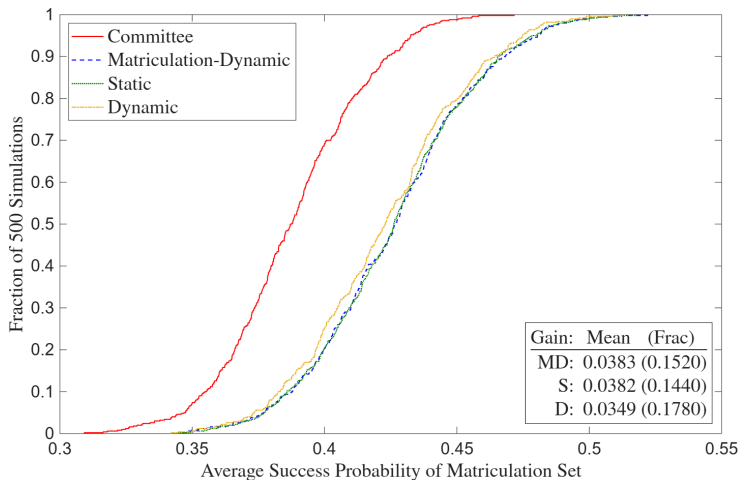
Q1s Puzos LL AC	2029	588	3408.8	29837.8
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►► Restricted Matriculation-Dynamic Estimates

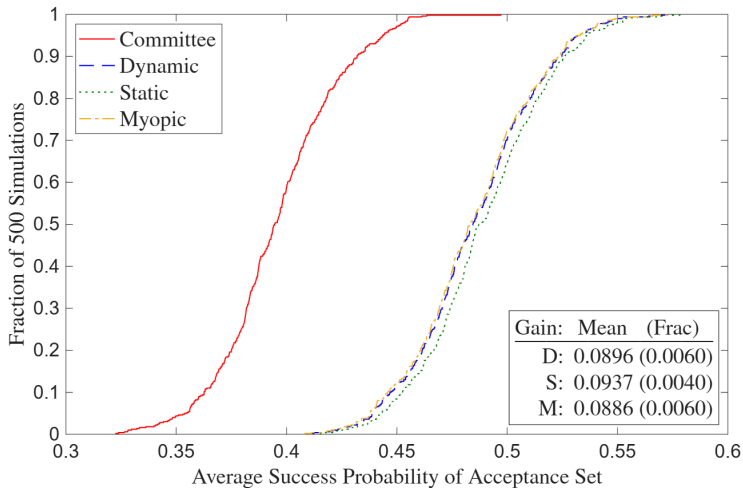
A3: Economics Department Deviation Gains (Acceptance)



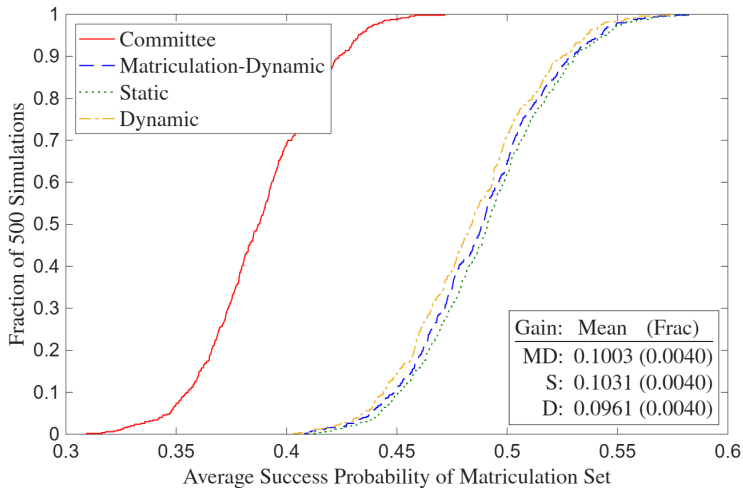
A3: Economics Department Deviation Gains (Matriculation)



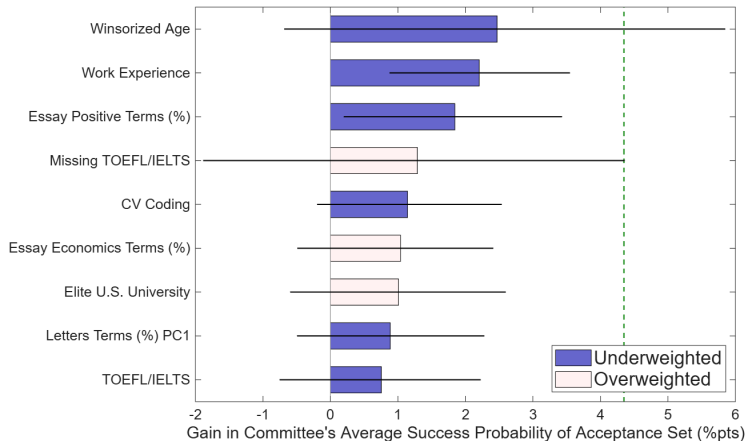
A3: Uncorrected Deviation Gains (Acceptance)



A3: Uncorrected Deviation Gains (Matriculation)

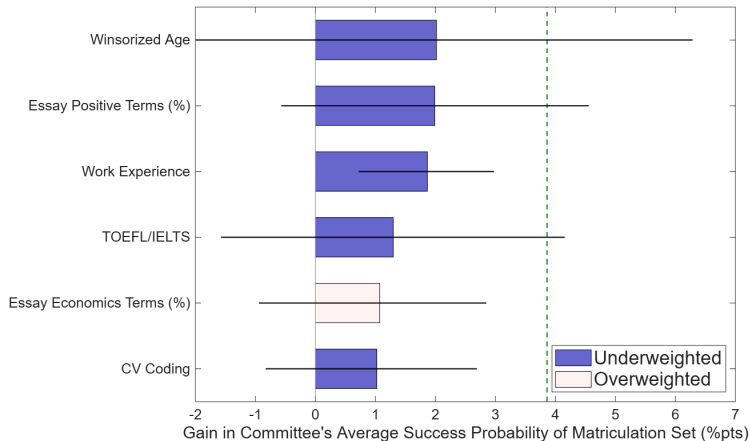


A3: Economics Department Gains Attribution (Acceptance)



- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

A3: Economics Department Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

A3: Economics Department Static Attribution

Table 12: Economics Department Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.4692	25.9910	26.2840	29.9572	29.5331	1.1674	1.0326
Work Experience	2.2037	0.5247	0.7276	0.9844	0.9650	0.5141	0.4752
Essay Positive Terms (%)	1.8431	11.7499	11.6521	12.5376	12.1183	0.7858	0.4137
Missing TOEFL/IELTS	1.2872	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.1426	2.8001	2.7820	3.3399	2.9415	0.6244	0.1785
Essay Economics Terms (%)	1.0382	9.7228	9.7423	9.0284	9.5388	-0.6411	-0.1828
Elite U.S. University	1.0086	0.1267	0.2335	0.0078	0.0389	-0.6784	-0.5848
Letters Terms (%) PC1	0.8868	-0.0106	0.1374	0.4969	0.1387	0.5202	0.0018
TOEFL/IELTS	0.7541	7.4376	7.5675	7.7759	7.6827	0.5957	0.3294

» Deviation Gains Attribution

A3: Economics Department Matriculation-Static Attribution

Table 13: Economics Department Matriculation-Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.0176	25.9910	26.3284	30.9159	29.7737	1.4580	1.0950
Essay Positive Terms (%)	1.9874	11.7499	11.6107	12.9136	12.2763	1.1534	0.5905
Work Experience	1.8659	0.5247	0.7938	0.9949	0.9799	0.4077	0.3725
TOEFL/IELTS	1.2973	7.4376	7.4939	7.8501	7.6128	1.0238	0.3397
Essay Economics Terms (%)	1.0726	9.7228	9.9994	9.0856	9.5738	-0.8271	-0.3821
CV Coding	1.0219	2.8001	2.6661	3.3395	2.8782	0.7618	0.2374

►► Deviation Gains Attribution (Matriculation)

A3: Deviation Gains Robustness

	Acceptance Set	Matriculation Set
Observed Gain, Uncorrected	0.1015	0.1124
Observed Gain, Corrected	0.0514	0.0475
Perturbed Gain, Uncorrected	0.0937	0.1031
Perturbed Gain, Corrected	0.0435	0.0382
Likelihood-Ratio P-Value	0.1166	0.1166
Calibrated Signal Scale	0.4197	0.4197
Calibrated In-Sample Gain	0.1242	0.1481
Calibrated True Gain	0.0491	0.0560

- 1 Scale down (x, z) parameters with values of scaling factor τ from 0 to 1
- 2 Simulate *Success*, and find specific τ yielding observed joint LR test statistic
- 3 Calculate true deviation gain for that τ

» Deviation Gains (Acceptance)

» Deviation Gains (Matriculation)

A3: Graduate School Means by Program

Table 14a: Graduate School Means by Program

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

A3: Graduate School Means by Program, Accept = 1

Table 14b: Graduate School Means by Program, Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

A3: Graduate School Correlation Matrix

Table 15: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREa	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

A3: Graduate School Logistic Regression Results

Table 16a: Graduate School Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966** (0.2010)	-0.4332* (0.2612)	-0.5792** (0.3470)	-0.5501 (0.4120)	-0.1766 (0.5545)
Winsorized Age	-0.0995*** (0.0358)	0.0454* (0.0271)	0.0380 (0.0327)	0.0222 (0.0532)	-0.0520 (0.0890)
Female	0.4079** (0.1681)	0.6721*** (0.2299)	0.6539** (0.2971)	0.1311 (0.3452)	-0.5660 (0.5606)
U.S. African/Hispanic/Native	0.4221 (0.5258)	1.0081*** (0.3545)	1.1601*** (0.4426)	0.0120 (0.8411)	-0.8263 (1.1511)
U.S. Asian	-0.5394 (0.4190)	-0.5660 (0.6504)	-0.3260 (1.2512)	0.8638 (0.7201)	1.1711 (0.7276)
International Asian	-0.8445*** (0.2945)	-0.1853 (0.4146)	0.9405 (0.7233)	0.4890 (0.6729)	-0.7827 (0.8323)
International Other	0.5034 (0.3172)	0.3781 (0.4131)	1.3529*** (0.5086)	-0.3061 (0.7483)	0.6239 (1.1070)
GRE Verbal Percentile	0.0155*** (0.0050)	0.0281* (0.0160)	0.0753** (0.0344)	-0.0092 (0.0154)	0.9607*** (0.0218)
GRE Quantitative Percentile	0.1547*** (0.0186)	0.0161* (0.0086)	0.0219* (0.0116)	0.0211 (0.0150)	-0.0138 (0.0159)
GRE Analytic Percentile	0.0179*** (0.0042)	0.0199** (0.0094)	0.0030 (0.0134)	0.0175 (0.0127)	-0.0202 (0.0218)
Undergraduate GPA	2.2212*** (0.7219)	0.6969 (0.5476)	1.3849** (0.5513)	1.2646 (0.9703)	1.7722* (0.9487)
TOEFL/IELTS	0.3878 (0.2883)	0.6456 (0.4708)	1.0963** (0.5526)	0.9655* (0.5496)	-0.4656 (0.6112)
Elite U.S. University	0.8850*** (0.2960)	0.3943 (0.3135)	0.5541* (0.3802)	0.4657 (0.4085)	1.4303** (0.6102)
Elite U.S. Liberal Arts	1.1173*** (0.3813)	0.6279* (0.3754)	1.0698** (0.5190)	-0.6088 (1.4106)	2.3151** (1.0842)
Elite Foreign University	1.0061* (0.6050)	0.4759* (0.8224)	0.4759* (0.7938)	1.7718*** (0.8587)	1.6165* (1.9880)
Other Foreign University	0.5442 (0.5785)	0.5777 (0.7688)	1.3733** (0.6902)	0.4188 (0.7157)	-1.0737 (1.3016)
Graduate Degree	0.0467 (0.2238)	0.0733 (0.2583)	0.3128 (0.3280)	0.4849 (0.4896)	0.4892 (0.6390)
Work Experience	0.3932** (0.1889)	0.4027 (0.2993)	0.4972 (0.3265)	0.1495 (0.3778)	1.8201** (0.8564)
#Professor Recommenders	0.0346 (0.1317)	-0.0178 (0.1339)	0.5511*** (0.2554)	0.4701* (0.2652)	-0.0981 (0.2660)
#Prolific Recommenders	0.2635*** (0.0854)	0.3025** (0.1262)	0.1880 (0.1576)	0.3284** (0.1544)	0.5001* (0.3145)
#Math Courses	0.1689*** (0.0596)				
Advanced Math	0.2319 (0.1790)				
Missing GRE	0.1899 (0.7646)	-0.3259 (0.3100)	0.6282 (0.4275)	-0.0831 (0.3761)	-1.1514** (0.5702)
Missing Undergraduate GPA	-0.4745 (0.4891)	-0.4017 (0.5440)	-1.5492** (0.6893)	-1.1087* (0.6401)	0.4890 (1.1831)
Missing TOEFL/IELTS	-0.3786 (0.2432)	0.6188 (0.5657)	0.6745 (0.5190)	-0.0440 (0.5099)	-1.7224** (0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2677	0.1525	0.1962	0.1871	0.2475

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive** ($p < .1$): Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative** ($p < .1$): Covid Year, Winsorized Age, International Asian

A3: Graduate School Logistic Regression AMEs

Table 16b: Graduate School Logistic Regression AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303** (0.0153)	-0.0164 (0.0300)	-0.0431** (0.0261)	-0.0298 (0.0224)	-0.0075 (0.0238)	-0.0218*** (0.0084)
Winsorized Age	0.0017** (0.0027)	0.0017** (0.0010)	0.0028 (0.0025)	0.0012 (0.0020)	-0.0022 (0.0038)	0.0016* (0.0009)
Female	0.0332*** (0.0129)	0.0254*** (0.0088)	0.0400*** (0.0220)	0.0071 (0.0187)	-0.0216 (0.0239)	0.0201*** (0.0075)
U.S. African/Hispanic/Native	0.0323 (0.0402)	0.0381*** (0.0137)	0.0089*** (0.0332)	0.0007 (0.0455)	-0.0352 (0.0487)	0.0384*** (0.0119)
U.S. Asian	-0.0413 (0.0322)	-0.0215 (0.0247)	-0.0244 (0.0936)	0.0468 (0.0391)	0.0099 (0.0513)	0.0062 (0.0185)
International Asian	-0.0646*** (0.0224)	-0.0070 (0.0157)	0.0704 (0.0543)	0.0265 (0.0363)	-0.0334 (0.0393)	0.0069 (0.0132)
International Other	0.0385 (0.0423)	0.0143 (0.0157)	0.0113*** (0.0385)	-0.0057 (0.0406)	0.0266 (0.0471)	0.0238* (0.0132)
GRE Verbal Percentile	0.0012*** (0.0005)	0.0011* (0.0006)	0.0056** (0.0025)	-0.0005 (0.0008)	0.0026*** (0.0010)	0.0015*** (0.0005)
GRE Quantitative Percentile	0.0118*** (0.0014)	0.0056** (0.0003)	0.0016* (0.0009)	0.0011 (0.0008)	-0.0006 (0.0007)	0.0016*** (0.0003)
GRE Analytic Percentile	0.0014*** (0.0003)	0.0008** (0.0004)	0.0002 (0.0010)	-0.0009 (0.0007)	-0.0009 (0.0009)	0.0007** (0.0003)
Undergraduate GPA	0.1699*** (0.0551)	0.0264 (0.0308)	0.0222** (0.0423)	0.0984 (0.0522)	0.0757** (0.0418)	0.0486*** (0.0176)
TOEFL/IELTS	0.0207 (0.0219)	0.0044 (0.0179)	0.0523** (0.0417)	-0.0198 (0.0301)	0.0326*** (0.0259)	0.0326*** (0.0123)
Elite U.S. University	0.0277*** (0.0225)	0.0149 (0.0119)	0.0410** (0.0281)	0.0610*** (0.0264)	-0.0233*** (0.0271)	0.0420*** (0.0098)
Elite U.S. Liberal Arts College	0.0288 (0.0288)	0.0238* (0.0143)	0.0801** (0.0402)	-0.0130 (0.0761)	0.0667** (0.0480)	0.0420*** (0.0136)
Elite Foreign University	0.0180 (0.0463)	0.0770** (0.0311)	0.0677** (0.0591)	0.0166 (0.0481)	0.0479*** (0.0650)	0.0063 (0.0241)
Other Foreign University	0.0416 (0.0441)	0.0219 (0.0291)	0.1028* (0.0527)	0.0227 (0.0391)	-0.0458 (0.0556)	0.0273 (0.0221)
Graduate Degree	0.0026 (0.0171)	0.0028 (0.0098)	0.0234 (0.0247)	0.0219 (0.0262)	0.0208 (0.0273)	0.0106 (0.0089)
Work Experience	0.0301** (0.0144)	0.0152 (0.0113)	0.0372 (0.0248)	0.0081 (0.0205)	0.0776** (0.0368)	0.0221** (0.0089)
#Professor Recommenders	0.0005 (0.0101)	-0.0007 (0.0051)	0.0209* (0.0190)	-0.0042 (0.0142)	0.0263 (0.0113)	0.0063 (0.0047)
#Prolific Recommenders	0.0202*** (0.0065)	0.0114** (0.0048)	0.0141 (0.0118)	0.0178** (0.0082)	0.0251* (0.0130)	0.0133*** (0.0038)
#Math Courses	0.0129*** (0.0045)					
Advanced Math	0.0177 (0.0137)					
Missing GRE	0.0145 (0.0585)	-0.0123 (0.0118)	0.0470 (0.0319)	-0.0045 (0.0204)	-0.0401* (0.0252)	-0.0062 (0.0088)
Missing Undergraduate GPA	-0.0363 (0.0375)	-0.0186 (0.0266)	-0.1160** (0.0524)	-0.0601* (0.0355)	0.0200 (0.0605)	-0.0291 (0.0179)
Missing TOEFL/IELTS	-0.0290 (0.0188)	0.0234 (0.0215)	0.0505 (0.0388)	-0.0034 (0.0276)	-0.0734** (0.0417)	0.0108 (0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	115.8637	23	0.0000			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

A3: Graduate School Logistic Regression Results (Synthetic)

Table 17a: Graduate School Logistic Regression Results (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997 (0.1633)	-0.0223 (0.2215)	0.2028 (0.3078)	-0.4791 (0.3477)	-0.5946 (0.4520)
Winsorized Age	-0.0509** (0.0201)	-0.0198 (0.0174)	0.0576* (0.0307)	-0.0164 (0.0357)	0.1203*** (0.0400)
Female	0.1941 (0.1402)	0.0923 (0.2004)	0.1389 (0.2526)	0.2595 (0.3255)	0.2119 (0.4310)
U.S. African/Hispanic/Native	0.3053 (0.3724)	-0.1457 (0.4305)	-0.2379 (0.4670)	-1.6401 (1.0720)	0.5413 (0.5306)
U.S. Asian	0.5663 (0.3642)	0.6016 (0.4452)	-0.7804 (1.1675)	-0.4305 (0.7036)	0.7122 (0.8863)
International Asian	-0.1428 (0.2169)	0.1630 (0.2803)	0.5780 (0.3740)	-0.4065 (0.4275)	0.1868 (0.6866)
International Other	-0.1231 (0.2111)	0.1395 (0.3046)	0.1097 (0.3237)	-0.1602 (0.3832)	-1.0807* (0.7583)
GRE Verbal Percentile	0.0167*** (0.0040)	0.0247** (0.0117)	0.0164 (0.0127)	-0.0029 (0.0100)	0.0132 (0.0184)
GRE Quantitative Percentile	0.0233*** (0.0090)	-0.0027 (0.0051)	0.0026 (0.0088)	0.0304** (0.0137)	0.0097 (0.0090)
GRE Analytic Percentile	0.0042 (0.0031)	0.0004 (0.0050)	-0.0073 (0.0092)	-0.0042 (0.0101)	0.0069 (0.0103)
Undergraduate GPA	0.2292 (0.4745)	0.6327 (0.4085)	1.3552*** (0.5796)	-1.3371** (0.5584)	-0.1002 (0.6152)
TOEFL/IELTS	-0.4673** (0.2157)	0.5141 (0.4185)	-0.7648 (0.6814)	0.2446 (0.7498)	-0.7354 (0.8025)
Elite U.S. University	-0.4850* (0.2560)	-0.2243 (0.3258)	0.0572 (0.3421)	0.4764 (0.4788)	0.3941 (0.4782)
Elite U.S. Liberal Arts College	0.2429 (0.3049)	0.1750 (0.4042)	0.2358 (0.4964)	0.4860 (0.9631)	0.9652 (0.7261)
Elite Foreign University	-0.1087 (0.3015)	-0.1622 (0.7484)	-0.1029 (0.5668)	0.4801 (0.7841)	3.0516*** (0.8673)
Other Foreign University	-0.1079 (0.1979)	0.2043 (0.2732)	0.2721 (0.3391)	0.1731 (0.4202)	0.3023 (0.5926)
Graduate Degree	-0.0186 (0.1624)	-0.0186 (0.2439)	-0.1661 (0.2765)	-0.0698*** (0.3320)	-0.2769 (0.3814)
Work Experience	0.2525* (0.1441)	-0.0391 (0.2396)	-0.0736 (0.2629)	-0.2650 (0.3417)	-1.1936** (0.4605)
#Professor Recommenders	-0.1059 (0.0909)	0.0911 (0.1138)	-0.1838 (0.1473)	0.2161 (0.1817)	-0.4924** (0.1956)
#Profic Recommenders	0.0039 (0.0636)	0.2189** (0.1108)	-0.1743 (0.1723)	0.0673 (0.1642)	0.5134** (0.2213)
#Math Courses	0.0396 (0.0421)				
Advanced Math	0.1961 (0.1416)				
Missing GRE	-0.1093 (0.4328)	-0.2029 (0.2195)	-0.0081 (0.2689)	-0.0433 (0.3503)	-0.6040 (0.4017)
Missing Undergraduate GPA	0.1531 (0.1540)	0.0895 (0.2293)	-0.2455 (0.3015)	0.6298** (0.3125)	-0.3574 (0.4342)
Missing TOEFL/IELTS	-0.2909* (0.1640)	0.4698 (0.3480)	0.5752 (0.5466)	0.1346 (0.4345)	-0.0781*** (0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2251	690	737	486
Pseudo-R ²	0.0524	0.0366	0.0753	0.1046	0.2230

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive** ($p < .1$): GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative** ($p < .1$): Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A3: Graduate School Logistic Regression AMEs (Synthetic)

Table 17b: Graduate School Logistic Regression AMEs (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204 (0.0167)	-0.0014 (0.0099)	0.0192 (0.0296)	-0.0298 (0.0216)	-0.0345 (0.0262)	-0.0089 (0.0086)
Winsorized Age	-0.0052** (0.0021)	-0.0009 (0.0038)	0.0054** (0.0029)	-0.0010 (0.0022)	0.0070*** (0.0023)	0.0004 (0.0006)
Female	0.0198 (0.0143)	0.0041 (0.0089)	0.0162 (0.0239)	0.0123 (0.0204)	0.0099 (0.0251)	0.0099 (0.0078)
U.S. African/Hispanic/Native	0.0312 (0.0381)	-0.0065 (0.0192)	-0.0225 (0.0441)	-0.1021 (0.0673)	0.0314 (0.0396)	-0.0137 (0.0147)
U.S. Asian	0.0679 (0.0372)	0.0266 (0.0199)	-0.0750 (0.1105)	-0.0268 (0.0495)	0.0413 (0.0516)	0.0241 (0.0179)
International Asian	-0.0146 (0.0222)	0.0073 (0.0125)	0.0546 (0.0353)	-0.0309 (0.0263)	0.0108 (0.0399)	0.0075 (0.0112)
International Other	-0.0126 (0.0236)	0.0062 (0.0136)	0.0104 (0.0306)	-0.0100 (0.0237)	-0.0027 (0.0443)	-0.0014 (0.0106)
GRE Verbal Percentile	0.0017*** (0.0005)	0.0011** (0.0005)	0.0015 (0.0012)	-0.0002 (0.0006)	0.0008 (0.0010)	0.0008** (0.0004)
GRE Quantitative Percentile	0.0024*** (0.0009)	-0.0001 (0.0002)	0.0003 (0.0008)	0.0019*** (0.0009)	0.0006 (0.0006)	0.0003 (0.0002)
GRE Analytic Percentile	0.0004 (0.0003)	0.0000 (0.0003)	-0.0007 (0.0009)	-0.0003 (0.0006)	0.0004 (0.0006)	0.0000 (0.0002)
Undergraduate GPA	0.0234 (0.0485)	0.0282 (0.0223)	0.1281*** (0.0553)	0.0620** (0.0358)	0.0016 (0.0357)	0.0130 (0.0158)
TOEFL/IELTS	-0.0478** (0.0221)	0.0229 (0.0187)	-0.0723 (0.0645)	0.0152 (0.0466)	-0.0427 (0.0469)	0.0050 (0.0187)
Elite U.S. University	-0.0466* (0.0261)	-0.0100 (0.0145)	0.0207 (0.0324)	0.0297 (0.0297)	0.0229 (0.0274)	0.0036 (0.0108)
Elite U.S. Liberal Arts College	0.0248 (0.0312)	0.0078 (0.0180)	0.0223 (0.0469)	0.0303 (0.0602)	0.0502 (0.0410)	0.0217 (0.0160)
Elite Foreign University	-0.0166 (0.0308)	-0.0512 (0.0337)	0.0700 (0.0537)	0.0299 (0.0400)	0.1771*** (0.0484)	0.0156 (0.0179)
Other Foreign University	-0.0111 (0.0202)	0.0091 (0.0122)	0.0257 (0.0321)	0.0108 (0.0267)	0.0175 (0.0344)	0.0116 (0.0103)
Graduate Degree	-0.0020 (0.0166)	0.0079 (0.0109)	-0.0015 (0.0261)	-0.0542*** (0.0206)	-0.0161 (0.0220)	-0.0063 (0.0085)
Work Experience	0.0258* (0.0146)	-0.0017 (0.0107)	-0.0070 (0.0248)	-0.0165 (0.0213)	-0.0692** (0.0275)	-0.0093 (0.0086)
#Professor Recommenders	-0.0108 (0.0093)	0.0041 (0.0051)	-0.0173 (0.0140)	0.0136 (0.0114)	-0.0265*** (0.0110)	-0.0027 (0.0042)
#Profic Recommenders	0.0004 (0.0097)	0.0096* (0.0050)	-0.0165 (0.0164)	0.0042 (0.0102)	0.0298** (0.0126)	0.0033* (0.0045)
#Math Courses	0.0037 (0.0043)	0.0037 (0.0043)				
Advanced Math	0.0201 (0.0145)					
Missing GRE	-0.0204 (0.0443)	-0.0090 (0.0098)	-0.0001 (0.0254)	-0.0027 (0.0218)	-0.0350 (0.0239)	-0.0008 (0.0079)
Missing Undergraduate GPA	0.0157 (0.0158)	0.0040 (0.0102)	-0.0232 (0.0285)	0.0392* (0.0201)	-0.0207 (0.0254)	0.0019 (0.0063)
Missing TOEFL/IELTS	-0.0305* (0.0168)	0.0218 (0.0156)	0.0544 (0.0517)	0.0684 (0.0272)	-0.1265*** (0.0434)	0.0063 (0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	37.4658	23	0.0260			

Robust standard errors in parentheses *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A3: Graduate School Lasso Logistic Regression Results

Table 18a: Graduate School Lasso Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Winsorized Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R ²	0.2297	0.0912	0.0592	0.0850	0.0537

- Economics Positive:** Female, U.S. African/Hispanic/Native, International Other, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, TOEFL/IELTS, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, # Prolific Recommenders, #Math Courses, Advanced Math
- Economics Negative:** Covid Year, Winsorized Age, U.S. Asian, International Asian, Missing TOEFL/IELTS

A4: Simulated Wald Test Results (Same)

Table 19a: Simulated Wald Test Results (Same)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3848*** (1.1315)	-1.5822*** (0.0956)	-28.5524*** (1.4963)	-1.3677*** (0.0500)	-10.4586*** (0.7236)	-1.9105*** (0.1070)	-10.2610*** (0.4815)	-1.8514*** (0.0628)
GRE Quantitative Percentile	0.1509*** (0.0131)	0.0146*** (0.0012)	0.3245*** (0.0179)	0.0155*** (0.0007)	0.1058*** (0.0088)	0.0193*** (0.0014)	0.1154*** (0.0070)	0.0208*** (0.0011)
Gender	0.4714*** (0.1475)	0.0455*** (0.0140)	0.7880*** (0.1696)	0.0377*** (0.0081)	-0.1384 (0.1095)	-0.0253 (0.0200)	-0.0497 (0.0895)	-0.0090 (0.0161)
Demographic 1	-0.6287*** (0.2088)	-0.0607*** (0.0200)	-1.8488*** (0.2445)	-0.0886*** (0.0115)	-0.3486** (0.1467)	-0.0637** (0.0267)	-0.3079*** (0.1123)	-0.0556*** (0.0202)
Demographic 2	0.1531 (0.2072)	0.0148 (0.0200)	0.3703* (0.1929)	0.0177* (0.0091)	0.1756 (0.1558)	0.0321 (0.0284)	0.0282 (0.1069)	0.0051 (0.0193)
Advanced Math	0.2889* (0.1612)	0.0279* (0.0156)			0.3267*** (0.1112)	0.0597*** (0.0202)		
Committee Score	0.2241*** (0.0413)	0.0216*** (0.0039)	0.3994*** (0.0506)	0.0191*** (0.0023)	0.1026*** (0.0287)	0.0187*** (0.0052)	0.0757*** (0.0228)	0.0137*** (0.0041)
Non-Economics 1			0.8376*** (0.2297)	0.0401*** (0.0107)			1.9483*** (0.1236)	0.3515*** (0.0205)
Observations Pseudo-R ²	5000	0.3251			5000	0.1431		
Parameters LL AIC	14	-1130.4	2288.8		14	-2691.4	5410.8	
Wald Statistic DF P-Value	3.1018	5	0.6843		3.0042	5	0.6993	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A4: Simulated Wald Test Results (Different)

Table 19b: Simulated Wald Test Results (Different)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3291*** (1.1274)	-1.5784*** (0.0955)	-8.4753*** (0.8673)	-0.3642*** (0.0412)	-10.4581*** (0.7237)	-1.9104*** (0.1070)	-3.2236*** (0.4018)	-0.5997*** (0.0728)
GRE Quantitative Percentile	0.1502*** (0.0131)	0.0145*** (0.0012)	0.0402*** (0.0128)	0.0017*** (0.0006)	0.1058*** (0.0088)	0.0193*** (0.0014)	0.0034 (0.0062)	0.0006 (0.0012)
Gender	0.4702*** (0.1473)	0.0454*** (0.0140)	0.5942*** (0.1820)	0.0255*** (0.0079)	-0.1372 (0.1095)	-0.0251 (0.0200)	0.0375 (0.0868)	0.0070 (0.0161)
Demographic 1	-0.6246*** (0.2085)	-0.0604*** (0.0200)	-0.2541 (0.2393)	-0.0109 (0.0103)	-0.3544** (0.1467)	-0.0647** (0.0267)	-0.1152 (0.1103)	-0.0214 (0.0205)
Demographic 2	0.1555 (0.2070)	0.0150 (0.0200)	0.0081 (0.2109)	0.0003 (0.0091)	0.1713 (0.1557)	0.0313 (0.0284)	-0.0562 (0.1030)	-0.0105 (0.0192)
Advanced Math	0.2886* (0.1611)	0.0279* (0.0156)			0.3284*** (0.1112)	0.0600*** (0.0202)		
Committee Score	0.2241*** (0.0413)	0.0217*** (0.0039)	0.3817*** (0.0528)	0.0164*** (0.0025)	0.1026*** (0.0287)	0.0187*** (0.0052)	0.1808*** (0.0228)	0.0336*** (0.0041)
Non-Economics 1			-0.0452 (0.1995)	-0.0019 (0.0086)			1.1393*** (0.1219)	0.2120*** (0.0219)
Observations Pseudo-R ²	5000	0.1488			5000	0.0758		
Parameters LL AIC	14	-1155.7	2339.5		14	-2747.2	5522.4	
Wald Statistic DF P-Value	112.5897	5	0.0000		124.7379	5	0.0000	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A4: Simulated Unrestricted Dynamic Estimates (Optimal)

Table 20a: Simulated Unrestricted Dynamic Estimates (Optimal)

	Advanced Math		Committee Score		Transition		Accept		Success	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577** (0.7805)	-0.4303** (0.1891)	0.3695 (1.4912)		-12.0690*** (0.9758)	-2.2483*** (0.1337)	-8.3663*** (1.7459)	-1.6540*** (0.3133)	-7.3172*** (1.2885)	-1.0451*** (0.1661)
GRE Quantitative Percentile	0.0251*** (0.0094)	0.0061*** (0.0023)	0.0510*** (0.0170)		0.1260*** (0.0114)	0.0235*** (0.0017)	0.0911*** (0.0200)	0.0180*** (0.0036)	0.1028*** (0.0142)	0.0147*** (0.0017)
Gender	-0.2435* (0.1312)	-0.0596* (0.0319)	0.1807 (0.2034)		0.0400 (0.1525)	0.0074 (0.0284)	0.0150 (0.2415)	0.0030 (0.0477)	0.2259 (0.1762)	0.0323 (0.0251)
Demographic 1	-0.0504 (0.1730)	-0.0123 (0.0424)	-0.6989** (0.2889)		0.0163 (0.2031)	0.0030 (0.0378)	-0.0503 (0.3722)	-0.0100 (0.0736)	-0.4384* (0.2358)	-0.0626* (0.0334)
Demographic 2	-0.3121 (0.1930)	-0.0764 (0.0470)	-0.4226 (0.3254)		0.1678 (0.2315)	0.0313 (0.0431)	-0.2878 (0.4145)	-0.0569 (0.0818)	-0.4682* (0.2734)	-0.0669* (0.0387)
Advanced Math			1.1347*** (0.1992)				0.2066 (0.2496)	0.0408 (0.0491)	0.7127*** (0.1791)	0.1018*** (0.0247)
Committee Score							0.0600 (0.0639)	0.0119 (0.0126)	-0.0790 (0.0688)	-0.0113 (0.0098)
Missing Committee Score									0.0486 (0.1969)	0.0069 (0.0281)
Year Transition More	-0.2523** (0.1282)	-0.0618** (0.0311)	0.1483 (0.2025)		1.1441*** (0.1501)	0.2131*** (0.0250)	-1.1114*** (0.2446)	-0.2197*** (0.0435)	-0.3543** (0.1794)	-0.0506** (0.0254)
Program Rank									-0.0302*** (0.0018)	-0.0043*** (0.0002)
$\sqrt{MSE}$			1.8393*** (0.0614)							
Obs Param LL AIC	1000	38	-2602.7		5281.5					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A4: Simulated Unrestricted Dynamic Estimates (Suboptimal)

Table 20b: Simulated Unrestricted Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score		Transition		Accept		Success	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577** (0.7805)	-0.4303** (0.1891)	1.2187 (1.4016)		-10.3454*** (0.9474)	-2.0151*** (0.1440)	-4.8695*** (1.5490)	-1.0694*** (0.3252)	-7.3017*** (1.2721)	-1.0448*** (0.1635)
GRE Quantitative Percentile	0.0251*** (0.0094)	0.0061*** (0.0023)	0.0431*** (0.0164)		0.1103*** (0.0111)	0.0215*** (0.0018)	0.0488*** (0.0180)	0.0107*** (0.0038)	0.1021*** (0.0140)	0.0146*** (0.0017)
Gender	-0.2435* (0.1312)	-0.0596* (0.0319)	0.1707 (0.2037)		-0.0472 (0.1485)	-0.0092 (0.0289)	0.1746 (0.2312)	0.0383 (0.0506)	0.2087 (0.1755)	0.0299 (0.0251)
Demographic 1	-0.0504 (0.1730)	-0.0123 (0.0424)	-0.7422*** (0.2732)		-0.4655** (0.1978)	-0.0907** (0.0382)	-0.5625* (0.3192)	-0.1235* (0.0689)	-0.4120* (0.2351)	-0.0590* (0.0334)
Demographic 2	-0.3121 (0.1930)	-0.0764 (0.0470)	-0.4868* (0.2925)		0.1277 (0.2186)	0.0249 (0.0426)	-0.1757 (0.3381)	-0.0386 (0.0742)	-0.3798 (0.2696)	-0.0543 (0.0384)
Advanced Math			1.1482*** (0.1999)				-0.0098 (0.2366)	-0.0021 (0.0520)	0.7178*** (0.1795)	0.1027*** (0.0248)
Committee Score							0.1714*** (0.0604)	0.0376*** (0.0128)	-0.0769 (0.0684)	-0.0110 (0.0098)
Missing Committee Score									0.1129 (0.1909)	0.0162 (0.0273)
Year Transition More	-0.2523** (0.1282)	-0.0618** (0.0311)	0.0871 (0.2014)		1.0879*** (0.1466)	0.2119*** (0.0255)	-0.7286*** (0.2324)	-0.1600*** (0.0485)	-0.3741** (0.1781)	-0.0535** (0.0253)
Program Rank									-0.0309*** (0.0019)	-0.0044*** (0.0002)
$\sqrt{MSE}$			1.8362*** (0.0603)							
Obs Param LL AIC	1000	38	-2648.4		5372.9					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A4: Simulated Restricted Dynamic Estimates (Optimal)

Table 21a: Simulated Restricted Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-1.9394** (0.7791)	0.3771 (1.4942)	-0.1457** (0.0663)	0.8644 (0.7591, 0.9843)	0.2458 (0.2300)	56.11% (44.89%, 66.74%)	-7.8112*** (1.1221)	
GRE Quantitative Percentile	0.0273*** (0.0094)	0.0509*** (0.0171)					0.1044*** (0.0128)	
Gender	-0.2478* (0.1310)	0.1808 (0.2034)					0.0937 (0.1163)	
Demographic 1	-0.0432 (0.1727)	-0.6993** (0.2904)					-0.2023 (0.1559)	
Demographic 2	-0.3026 (0.1930)	-0.4231 (0.3260)					-0.2735 (0.1817)	
Advanced Math		1.1347*** (0.1992)					0.4597*** (0.1452)	
Committee Score							-0.0021 (0.0436)	
Missing Committee Score							0.0240 (0.1891)	
Year Transition More	-0.2448* (0.1281)	0.1481 (0.2023)	-0.1684*** (0.0637)	0.7304 (0.6462, 0.8256)	0.4096 (0.2744)	65.82% (57.07%, 73.62%)	-0.3690** (0.1763)	
Program Rank							-0.0300*** (0.0018)	
$\sqrt{\text{MSE}}$		1.8393*** (0.0614)						
Sigma								0.1647*** (0.0242)
Obs Param LL AIC	1000	29	-2642.8	5343.7				

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A4: Simulated Restricted Dynamic Estimates (Suboptimal)

Table 21b: Simulated Restricted Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-1.9242** (0.7773)	1.0267 (1.4057)	-0.0633 (0.0892)	0.9387 (0.7882, 1.1179)	0.1263 (0.2508)	53.15% (40.97%, 64.97%)	-7.7219*** (1.1783)	
GRE Quantitative Percentile	0.0271*** (0.0093)	0.0453*** (0.0165)					0.1006*** (0.0134)	
Gender	-0.2513* (0.1310)	0.1636 (0.2039)					0.1366 (0.1297)	
Demographic 1	-0.0530 (0.1728)	-0.7462*** (0.2730)					-0.5186*** (0.1682)	
Demographic 2	-0.3004 (0.1928)	-0.4758 (0.2928)					-0.2123 (0.1992)	
Advanced Math		1.1476*** (0.1999)					0.4443*** (0.1595)	
Committee Score							0.0635 (0.0480)	
Missing Committee Score							0.1011 (0.1862)	
Year Transition More	-0.2466* (0.1281)	0.0910 (0.2010)	-0.2559*** (0.0710)	0.7268 (0.6177, 0.8551)	0.3083 (0.3058)	60.70% (50.39%, 70.13%)	-0.3596** (0.1749)	
Program Rank							-0.0303*** (0.0018)	
$\sqrt{\text{MSE}}$		1.8362*** (0.0603)						
Sigma								0.2258*** (0.0411)
Obs Param LL AIC	1000	29	-2686.1	5430.2				

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A4: Simulated Unrestricted Static Estimates (Optimal)

Table 22a: Simulated Unrestricted Static Estimates (Optimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-12.0690*** (0.9758)	-2.2483*** (0.1337)	-8.3663*** (1.7459)	-1.6540*** (0.3133)	-7.3875*** (1.1197)	-1.0821*** (0.1406)	-7.3172*** (1.2885)	-1.0451*** (0.1661)
GRE Quantitative Percentile	0.1260*** (0.0114)	0.0235*** (0.0017)	0.0911*** (0.0200)	0.0180*** (0.0036)	0.1043*** (0.0134)	0.0153*** (0.0016)	0.1028*** (0.0142)	0.0147*** (0.0017)
Gender	0.0400 (0.1525)	0.0074 (0.0284)	0.0150 (0.2415)	0.0030 (0.0477)	0.1580 (0.1732)	0.0231 (0.0254)	0.2259 (0.1762)	0.0323 (0.0251)
Demographic 1	0.0163 (0.2031)	0.0030 (0.0378)	-0.0503 (0.3722)	-0.0100 (0.0736)	-0.4556** (0.2269)	-0.0667** (0.0328)	-0.4384* (0.2358)	-0.0626* (0.0334)
Demographic 2	0.1678 (0.2315)	0.0313 (0.0431)	-0.2878 (0.4145)	-0.0569 (0.0818)	-0.5153* (0.2641)	-0.0755** (0.0383)	-0.4682* (0.2734)	-0.0669* (0.0387)
Advanced Math			0.2066 (0.2496)	0.0408 (0.0491)			0.7127*** (0.1791)	0.1018*** (0.0247)
Committee Score			0.0600 (0.0639)	0.0119 (0.0126)			-0.0790 (0.0688)	-0.0113 (0.0098)
Missing Committee Score							0.0486 (0.1969)	0.0069 (0.0281)
Year Transition More	1.1441*** (0.1501)	0.2131*** (0.0250)	-1.1114*** (0.2446)	-0.2197*** (0.0435)	-0.3942** (0.1689)	-0.0577** (0.0244)	-0.3543** (0.1794)	-0.0506** (0.0254)
Program Rank					-0.0299*** (0.0018)	-0.0044*** (0.0002)	-0.0302*** (0.0018)	-0.0043*** (0.0002)
Obs Param LL AIC	1000	31	-1632.5	3327.1				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A4: Simulated Unrestricted Static Estimates (Suboptimal)

Table 22b: Simulated Unrestricted Static Estimates (Suboptimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-10.3454*** (0.9474)	-2.0151*** (0.1440)	-4.8695*** (1.5490)	-1.0694*** (0.3252)	-7.2362*** (1.1209)	-1.0632*** (0.1418)	-7.3017*** (1.2721)	-1.0448*** (0.1635)
GRE Quantitative Percentile	0.1103*** (0.0111)	0.0215*** (0.0018)	0.0488*** (0.0180)	0.0107*** (0.0038)	0.1026*** (0.0134)	0.0151*** (0.0016)	0.1021*** (0.0140)	0.0146*** (0.0017)
Gender	-0.0472 (0.1485)	-0.0092 (0.0289)	0.1746 (0.2312)	0.0383 (0.0506)	0.1408 (0.1727)	0.0207 (0.0254)	0.2087 (0.1755)	0.0299 (0.0251)
Demographic 1	-0.4655** (0.1978)	-0.0907** (0.0382)	-0.5625* (0.3192)	-0.1235* (0.0689)	-0.4199* (0.2285)	-0.0617* (0.0332)	-0.4120* (0.2351)	-0.0590* (0.0334)
Demographic 2	0.1277 (0.2186)	0.0249 (0.0426)	-0.1757 (0.3381)	-0.0386 (0.0742)	-0.4343* (0.2605)	-0.0638* (0.0380)	-0.3798 (0.2696)	-0.0543 (0.0384)
Advanced Math			-0.0098 (0.2366)	-0.0021 (0.0520)			0.7178*** (0.1795)	0.1027*** (0.0248)
Committee Score			0.1714*** (0.0604)	0.0376*** (0.0128)			-0.0769 (0.0684)	-0.0110 (0.0098)
Missing Committee Score							0.1129 (0.1909)	0.0162 (0.0273)
Year Transition More	1.0879*** (0.1466)	0.2119*** (0.0255)	-0.7286*** (0.2324)	-0.1600*** (0.0485)	-0.4350** (0.1689)	-0.0639*** (0.0244)	-0.3741** (0.1781)	-0.0535** (0.0253)
Program Rank					-0.0305*** (0.0019)	-0.0045*** (0.0002)	-0.0309*** (0.0019)	-0.0044*** (0.0002)
Obs Param LL AIC	1000	31	-1673.0	3408.1				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A4: Simulated Restricted Static Estimates (Optimal)

Table 23a: Simulated Restricted Static Estimates (Optimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	1.2114*** (0.2502)	77.05% (67.28%, 84.58%)	0.3765 (0.2666)	59.30% (46.35%, 71.08%)	-7.7989*** (1.0685)	-6.9427*** (1.2241)	
GRE Quantitative Percentile					0.1065*** (0.0128)	0.0957*** (0.0139)	
Gender					0.0694 (0.1077)	0.1374 (0.1373)	
Demographic 1					-0.1638 (0.1410)	-0.3053 (0.1944)	
Demographic 2					-0.0994 (0.1623)	-0.3969* (0.2189)	
Advanced Math						0.5127*** (0.1456)	
Committee Score						-0.0104 (0.0458)	
Missing Committee Score						-0.0145 (0.1924)	
Year Transition More	-1.3691*** (0.2738)	46.07% (40.01%, 52.24%)	0.6127* (0.3291)	72.89% (61.87%, 81.67%)	-0.3712** (0.1649)	-0.3819** (0.1814)	
Program Rank					-0.0298*** (0.0018)	-0.0300*** (0.0018)	
Sigma							0.2021*** (0.0252)
Obs Param LL AIC	1000	22	-1639.7	3323.5			

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A4: Simulated Restricted Static Estimates (Suboptimal)

Table 23b: Simulated Restricted Static Estimates (Suboptimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	1.5071*** (0.3814)	81.86% (68.13%, 90.50%)	0.2881 (0.2858)	57.15% (43.24%, 70.02%)	-7.9222*** (1.1110)	-6.4145*** (1.2821)	
GRE Quantitative Percentile					0.1102*** (0.0132)	0.0864*** (0.0144)	
Gender					0.0340 (0.1208)	0.2000 (0.1433)	
Demographic 1					-0.4496*** (0.1586)	-0.5033*** (0.1917)	
Demographic 2					-0.1170 (0.1795)	-0.3225 (0.2194)	
Advanced Math						0.4844*** (0.1548)	
Committee Score						0.0524 (0.0491)	
Missing Committee Score						0.0419 (0.1918)	
Year Transition More	-1.6799*** (0.3946)	45.69% (39.03%, 52.51%)	0.3660 (0.3462)	65.79% (53.86%, 76.01%)	-0.4035** (0.1655)	-0.3931** (0.1786)	
Program Rank					-0.0303*** (0.0019)	-0.0304*** (0.0018)	
Sigma							0.2573*** (0.0392)
Obs Param LL AIC	1000	22	-1687.8	3419.7			

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A4: Simulated Unrestricted Myopic Estimates (Optimal)

Table 24a: Simulated Unrestricted Myopic Estimates (Optimal)

	Advanced Math		Committee Score		Transition		Accept		Success	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.1765** (0.0898)	0.0439** (0.0222)	4.4154*** (0.1909)		-12.0690*** (0.9758)	-2.2483*** (0.1337)	-8.3663*** (1.7459)	-1.6540*** (0.3133)	-7.3172*** (1.2885)	-1.0451*** (0.1661)
GRE Quantitative Percentile					0.1260*** (0.0114)	0.0235*** (0.0017)	0.0911*** (0.0200)	0.0180*** (0.0036)	0.1028*** (0.0142)	0.0147*** (0.0017)
Gender					0.0400 (0.1525)	0.0074 (0.0284)	0.0150 (0.2415)	0.0030 (0.0477)	0.2259 (0.1762)	0.0323 (0.0251)
Demographic 1					0.0163 (0.2031)	0.0030 (0.0378)	-0.0503 (0.3722)	-0.0100 (0.0736)	-0.4384* (0.2358)	-0.0626* (0.0334)
Demographic 2					0.1678 (0.2315)	0.0313 (0.0431)	-0.2878 (0.4145)	-0.0569 (0.0818)	-0.4682* (0.2734)	-0.0669* (0.0387)
Advanced Math			1.1642*** (0.1981)				0.2066 (0.2496)	0.0408 (0.0491)	0.7127*** (0.1791)	0.1018*** (0.0247)
Committee Score							0.0600 (0.0639)	0.0119 (0.0126)	-0.0790 (0.0688)	-0.0113 (0.0098)
Missing Committee Score									0.0486 (0.1969)	0.0069 (0.0281)
Year Transition More	-0.2485** (0.1268)	-0.0618** (0.0313)	0.1348 (0.2040)		1.1441*** (0.1501)	0.2131*** (0.0250)	-1.1114*** (0.2446)	-0.2197*** (0.0435)	-0.3543** (0.1794)	-0.0506** (0.0254)
Program Rank									-0.0302*** (0.0018)	-0.0043*** (0.0002)
$\sqrt{MSE}$			1.8730*** (0.0618)							
Obs Param LL AIC	1000	30	-2617.5		5295.0					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A4: Simulated Unrestricted Myopic Estimates (Suboptimal)

Table 24b: Simulated Unrestricted Myopic Estimates (Suboptimal)

	Advanced Math		Committee Score		Transition		Accept		Success	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.1765** (0.0898)	0.0439** (0.0222)	4.5276*** (0.1915)		-10.3454*** (0.9474)	-2.0151*** (0.1440)	-4.8695*** (1.5490)	-1.0694*** (0.3252)	-7.3017*** (1.2721)	-1.0448*** (0.1635)
GRE Quantitative Percentile					0.1103*** (0.0111)	0.0215*** (0.0018)	0.0488*** (0.0180)	0.0107*** (0.0038)	0.1021*** (0.0140)	0.0146*** (0.0017)
Gender					-0.0472 (0.1485)	-0.0092 (0.0289)	0.1746 (0.2312)	0.0383 (0.0506)	0.2087 (0.1755)	0.0299 (0.0251)
Demographic 1					-0.4655** (0.1978)	-0.0907** (0.0382)	-0.5625* (0.3192)	-0.1235* (0.0689)	-0.4120* (0.2351)	-0.0590* (0.0334)
Demographic 2					0.1277 (0.2186)	0.0249 (0.0426)	-0.1757 (0.3381)	-0.0386 (0.0742)	-0.3798 (0.2696)	-0.0543 (0.0384)
Advanced Math			1.1789*** (0.1973)				-0.0098 (0.2366)	-0.0021 (0.0520)	0.7178*** (0.1795)	0.1027*** (0.0248)
Committee Score							0.1714*** (0.0604)	0.0376*** (0.0128)	-0.0769 (0.0684)	-0.0110 (0.0098)
Missing Committee Score									0.1129 (0.1909)	0.0162 (0.0273)
Year Transition More	-0.2485** (0.1268)	-0.0618** (0.0313)	0.0712 (0.2044)		1.0879*** (0.1466)	0.2119*** (0.0255)	-0.7286*** (0.2324)	-0.1600*** (0.0485)	-0.3741** (0.1781)	-0.0535** (0.0253)
Program Rank									-0.0309*** (0.0019)	-0.0044*** (0.0002)
$\sqrt{MSE}$			1.8655*** (0.0600)							
Obs Param LL AIC	1000	30	-2662.4		5384.8					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A4: Simulated Restricted Myopic Estimates (Optimal)

Table 25a: Simulated Restricted Myopic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.1822** (0.0898)	4.4131*** (0.1915)	-0.1459** (0.0669)	0.8642 (0.7581, 0.9852)	0.2450 (0.2303)	56.09% (44.86%, 66.74%)	-7.8232*** (1.1263)	
GRE Quantitative Percentile							0.1055*** (0.0128)	
Gender							0.0872 (0.1160)	
Demographic 1							-0.2089 (0.1545)	
Demographic 2							-0.2850 (0.1804)	
Advanced Math		1.1642*** (0.1982)					0.4385*** (0.1452)	
Committee Score							-0.0125 (0.0436)	
Missing Committee Score							0.0243 (0.1889)	
Year Transition More	-0.2418* (0.1267)	0.1344 (0.2039)	-0.1671*** (0.0640)	0.7312 (0.6463, 0.8273)	0.4173 (0.2757)	65.98% (57.16%, 73.81%)	-0.3689** (0.1764)	
Program Rank							-0.0300*** (0.0018)	
$\sqrt{\text{MSE}}$		1.8730*** (0.0618)						
Sigma								0.1652*** (0.0244)
Obs Param LL AIC	1000	21	-2658.4	5358.7				

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A4: Simulated Restricted Myopic Estimates (Suboptimal)

Table 25b: Simulated Restricted Myopic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.1801** (0.0898)	4.5328*** (0.1919)	-0.0597 (0.0912)	0.9420 (0.7878, 1.1264)	0.1255 (0.2525)	53.13% (40.87%, 65.03%)	-7.7935*** (1.1728)	
GRE Quantitative Percentile							0.1025*** (0.0132)	
Gender							0.1327 (0.1299)	
Demographic 1							-0.5335*** (0.1674)	
Demographic 2							-0.2291 (0.1982)	
Advanced Math		1.1784*** (0.1973)					0.4297*** (0.1610)	
Committee Score							0.0501 (0.0488)	
Missing Committee Score							0.1041 (0.1858)	
Year Transition More	-0.2436* (0.1267)	0.0726 (0.2043)	-0.2565*** (0.0714)	0.7289 (0.6175, 0.8605)	0.3183 (0.3098)	60.92% (50.44%, 70.47%)	-0.3616** (0.1752)	
Program Rank							-0.0303*** (0.0018)	
$\sqrt{\text{MSE}}$		1.8654*** (0.0599)						
Sigma								0.2283*** (0.0424)
Obs Param LL AIC	1000	21	-2701.0	5444.0				

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A4: Simulated Unrestricted Matriculation-Dynamic Estimates (Optimal)

Table 26a: Simulated Unrestricted Matriculation-Dynamic Estimates (Optimal)

	Advanced Math		Committee Score		Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577** (0.7805)	-0.4303** (0.1891)	0.3695 (1.4912)		-12.0690*** (0.9758)	-2.2483*** (0.1337)	-8.3663*** (1.7459)	-1.6540*** (0.3133)	-7.3172*** (1.2885)	-1.0451*** (0.1661)	10.8487* (5.7339)	1.1240** (0.5653)
GRE Quantitative Percentile	0.0251*** (0.0094)	0.0061*** (0.0023)	0.0510*** (0.0170)		0.1260*** (0.0114)	0.0235*** (0.0017)	0.0911*** (0.0200)	0.0180*** (0.0036)	0.1028*** (0.0142)	0.0147*** (0.0017)	-0.1049* (0.0630)	-0.0109* (0.0063)
Gender	-0.2435* (0.1312)	-0.0596* (0.0319)	0.1807 (0.2034)		0.0400 (0.1525)	0.0074 (0.0284)	0.0150 (0.2415)	0.0030 (0.0477)	0.2259 (0.1762)	0.0323 (0.0251)	-0.9225 (0.5834)	-0.0956* (0.0575)
Demographic 1	-0.0504 (0.1730)	-0.0123 (0.0424)	-0.6989** (0.2889)		0.0163 (0.2031)	0.0030 (0.0378)	-0.0503 (0.3722)	-0.0100 (0.0736)	-0.4384* (0.2358)	-0.0626* (0.0334)	1.2693 (0.8109)	0.1315 (0.0828)
Demographic 2	-0.3121 (0.1930)	-0.0764 (0.0470)	-0.4226 (0.3254)		0.1678 (0.2315)	0.0313 (0.0431)	-0.2878 (0.4145)	-0.0569 (0.0818)	-0.4682* (0.2734)	-0.0669* (0.0387)	0.2958 (0.9734)	0.0306 (0.1014)
Advanced Math			1.1347*** (0.1992)				0.2066 (0.2496)	0.0408 (0.0491)	0.7127*** (0.1791)	0.1018*** (0.0247)	0.3819 (0.5414)	0.0396 (0.0557)
Committee Score							0.0600 (0.0639)	0.0119 (0.0126)	-0.0790 (0.0688)	-0.0113 (0.0098)	-1.0758*** (0.2376)	-0.1115*** (0.0153)
Missing Committee Score									0.0486 (0.1969)	0.0069 (0.0281)		
Year Transition/Matriculate More	-0.2523** (0.1282)	-0.0618** (0.0311)	0.1483 (0.2025)		1.1441*** (0.1501)	0.2131*** (0.0250)	-1.1114*** (0.2446)	-0.2197*** (0.0435)	-0.3543** (0.1794)	-0.0506** (0.0254)	5.2621*** (0.8695)	0.5452*** (0.0260)
Program Rank									-0.0302*** (0.0018)	-0.0043*** (0.0002)		
$\sqrt{MSE}$			1.8393*** (0.0614)									
Obs Param LL AIC	1000	46	-2644.2		5380.5							

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A4: Simulated Unrestricted Matriculation-Dynamic Estimates (Suboptimal)

Table 26b: Simulated Unrestricted Matriculation-Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score		Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577** (0.7805)	-0.4303** (0.1891)	1.2187 (1.4016)		-10.3454*** (0.9474)	-2.0151*** (0.1440)	-4.8695*** (1.5490)	-1.0694*** (0.3252)	-7.3017*** (1.2721)	-1.0448*** (0.1635)	5.7452 (4.2604)	0.5714 (0.4164)
GRE Quantitative Percentile	0.0251*** (0.0094)	0.0061*** (0.0023)	0.0431*** (0.0164)		0.1103*** (0.0111)	0.0215*** (0.0018)	0.0488*** (0.0180)	0.0107*** (0.0038)	0.1021*** (0.0140)	0.0146*** (0.0017)	-0.0443 (0.0468)	-0.0044 (0.0046)
Gender	-0.2435* (0.1312)	-0.0596* (0.0319)	0.1707 (0.2037)		-0.0472 (0.1485)	-0.0092 (0.0289)	0.1746 (0.2312)	0.0383 (0.0506)	0.2087 (0.1755)	0.0299 (0.0251)	-1.0062* (0.5693)	-0.1001* (0.0535)
Demographic 1	-0.0504 (0.1730)	-0.0123 (0.0424)	-0.7422*** (0.2732)		-0.4655** (0.1978)	-0.0907** (0.0382)	-0.5625* (0.3192)	-0.1235* (0.0689)	-0.4120* (0.2351)	-0.0590* (0.0334)	0.8111 (0.7062)	0.0807 (0.0712)
Demographic 2	-0.3121 (0.1930)	-0.0764 (0.0470)	-0.4868* (0.2925)		0.1277 (0.2186)	0.0249 (0.0426)	-0.1757 (0.3381)	-0.0386 (0.0742)	-0.3798 (0.2696)	-0.0543 (0.0384)	0.7188 (0.7190)	0.0715 (0.0724)
Advanced Math			1.1482*** (0.1999)				-0.0098 (0.2366)	-0.0021 (0.0520)	0.7178*** (0.1795)	0.1027*** (0.0248)	-0.7520 (0.5452)	-0.0748 (0.0534)
Committee Score							0.1714*** (0.0604)	0.0376*** (0.0128)	-0.0769 (0.0684)	-0.0110 (0.0098)	-1.0032*** (0.2212)	-0.0998*** (0.0150)
Missing Committee Score									0.1129 (0.1909)	0.0162 (0.0273)		
Year Transition/Matriculate More	-0.2523** (0.1282)	-0.0618** (0.0311)	0.0871 (0.2014)		1.0879*** (0.1466)	0.2119*** (0.0255)	-0.7286*** (0.2324)	-0.1600*** (0.0485)	-0.3741** (0.1781)	-0.0535** (0.0253)	5.6048*** (0.9335)	0.5574*** (0.0301)
Program Rank									-0.0309*** (0.0019)	-0.0044*** (0.0002)		
$\sqrt{MSE}$			1.8362*** (0.0603)									
Obs Param LL AIC	1000	46	-2693.0		5478.1							

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A4: Simulated Restricted Matriculation-Dynamic Estimates (Optimal)

Table 27a: Simulated Restricted Matriculation-Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-1.8113** (0.7828)	1.2405 (1.5031)	-1.4922*** (0.1913)	0.2249 (0.1546, 0.3272)	0.3506 (0.2619)	58.68% (45.94%, 70.35%)	-7.7566*** (1.1350)	2.9160 (5.1107)	
GRE Quantitative Percentile	0.0257*** (0.0094)	0.0408** (0.0173)					0.1057*** (0.0129)	-0.0133 (0.0562)	
Gender	-0.2464* (0.1312)	0.1731 (0.2034)					0.1159 (0.1353)	-0.9424* (0.5543)	
Demographic 1	-0.0487 (0.1729)	-0.7009** (0.2890)					-0.2877 (0.1882)	1.2129 (0.8220)	
Demographic 2	-0.3098 (0.1929)	-0.4291 (0.3258)					-0.3732* (0.2125)	0.5064 (0.9746)	
Advanced Math		1.1261*** (0.2002)					0.4974*** (0.1521)	0.3171 (0.5299)	
Committee Score							-0.0238 (0.0447)	-1.0550*** (0.2347)	
Missing Committee Score							0.0211 (0.1952)		
Year Transition/Matriculate More	-0.2441* (0.1282)	0.1538 (0.2051)	-1.2324*** (0.2881)	0.0656 (0.0355, 0.1212)	0.4094 (0.3147)	68.14% (57.83%, 76.93%)	-0.3789** (0.1823)	4.9595*** (0.9084)	
Program Rank							-0.0301*** (0.0018)		
$\sqrt{\text{MSE}}$		1.8440*** (0.0615)							
Sigma									0.1872*** (0.0329)
Obs Param LL AIC	1000	37	-2707.6	5489.2					

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

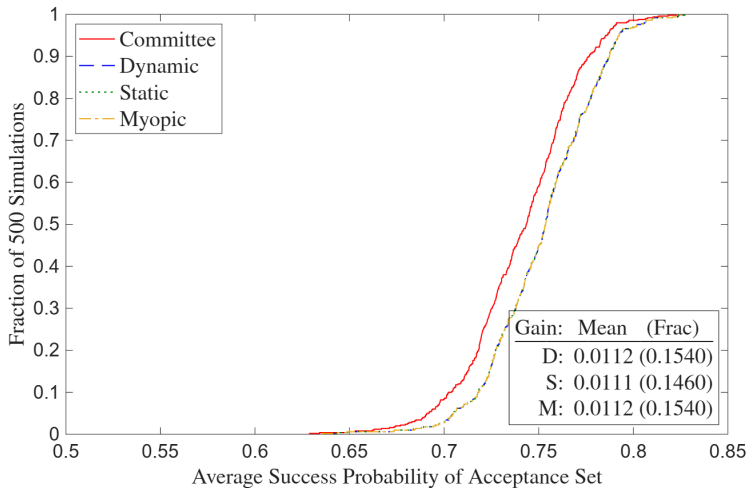
A4: Simulated Restricted Matriculation-Dynamic Estimates (Suboptimal)

Table 27b: Simulated Restricted Matriculation-Dynamic Estimates (Suboptimal)

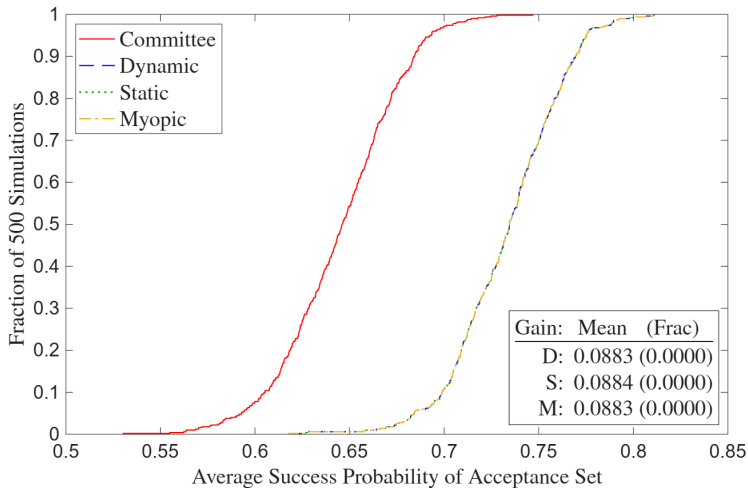
	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-1.7155** (0.7813)	1.8306 (1.4228)	-1.0818*** (0.2600)	0.3390 (0.2036, 0.5643)	0.2612 (0.3040)	56.49% (41.71%, 70.20%)	-7.7219*** (1.1701)	0.2437 (4.4569)	
GRE Quantitative Percentile	0.0246*** (0.0094)	0.0357** (0.0167)					0.1020*** (0.0133)	0.0200 (0.0495)	
Gender	-0.2456* (0.1312)	0.1683 (0.2035)					0.1877 (0.1499)	-1.0474* (0.5402)	
Demographic 1	-0.0469 (0.1731)	-0.7099*** (0.2735)					-0.5368*** (0.1983)	0.5144 (0.7157)	
Demographic 2	-0.3097 (0.1929)	-0.4873* (0.2925)					-0.3087 (0.2284)	0.7381 (0.7585)	
Advanced Math		1.1443*** (0.2001)					0.5177*** (0.1729)	-0.8396 (0.5352)	
Committee Score							0.0412 (0.0545)	-0.9473*** (0.2114)	
Missing Committee Score							0.0852 (0.1912)		
Year Transition/Matriculate More	-0.2478* (0.1281)	0.0862 (0.2022)	-1.0168*** (0.2514)	0.1226 (0.0646, 0.2327)	0.3707 (0.4109)	65.29% (50.97%, 77.29%)	-0.3780** (0.1794)	5.3618*** (0.9378)	
Program Rank							-0.0305*** (0.0018)		
$\sqrt{\text{MSE}}$		1.8366*** (0.0602)							
Sigma									0.2943*** (0.0717)
Obs Param LL AIC	1000	37	-2747.6	5569.2					

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

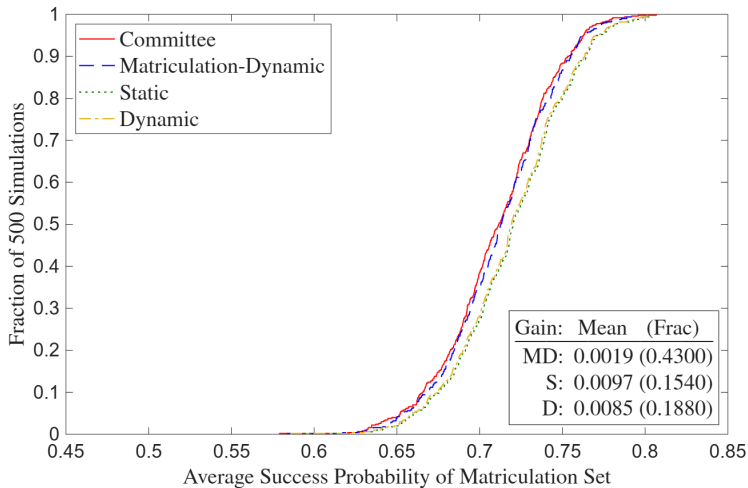
A4: Simulated Deviation Gains (Optimal)



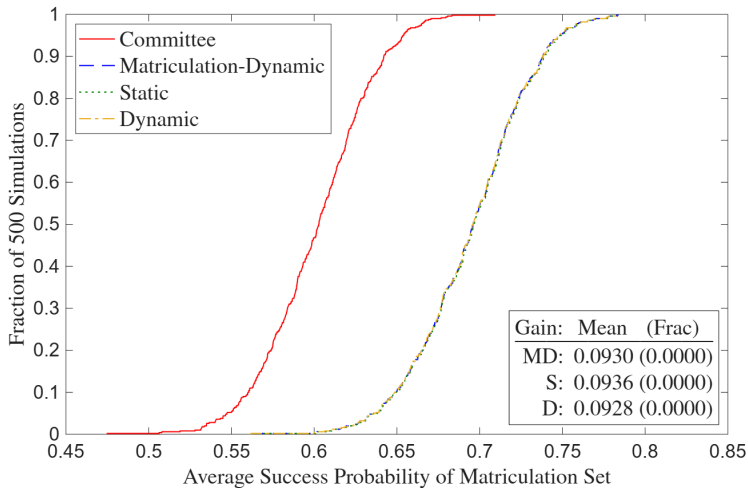
A4: Simulated Deviation Gains (Suboptimal)



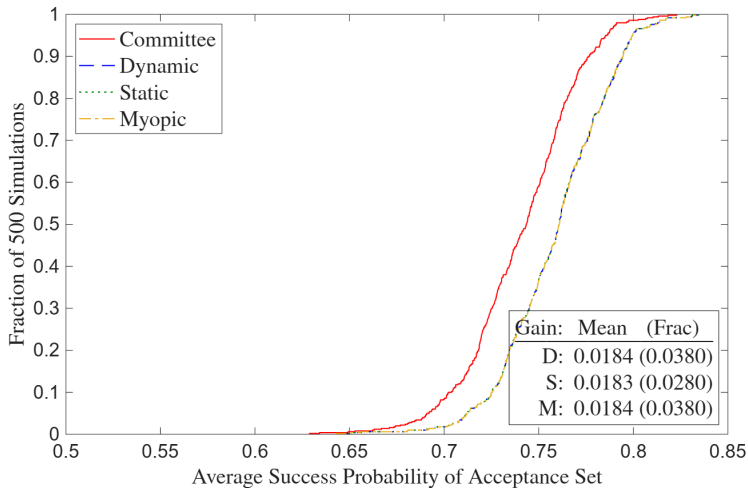
A4: Simulated Matriculation Deviation Gains (Optimal)



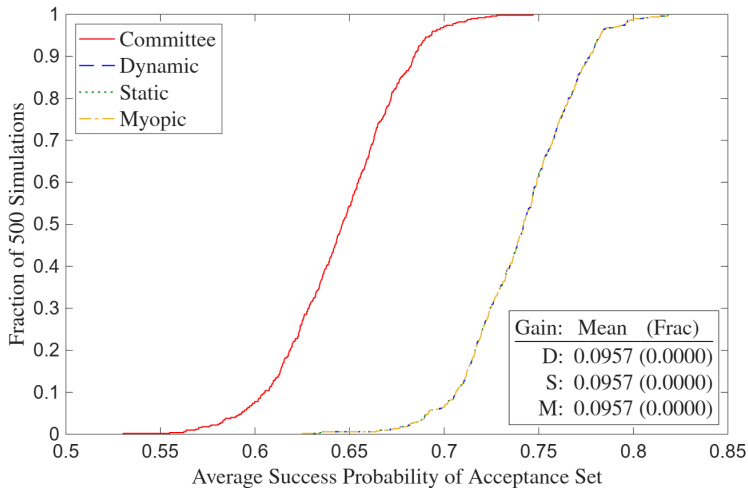
A4: Simulated Matriculation Deviation Gains (Suboptimal)



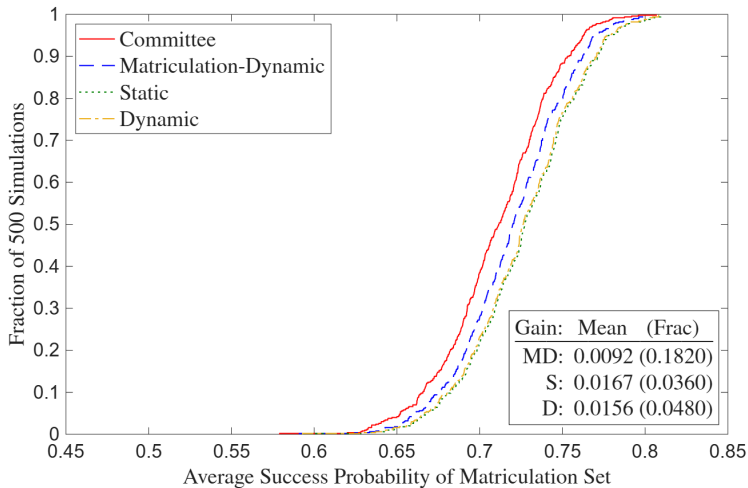
A4: Uncorrected Deviation Gains (Optimal)



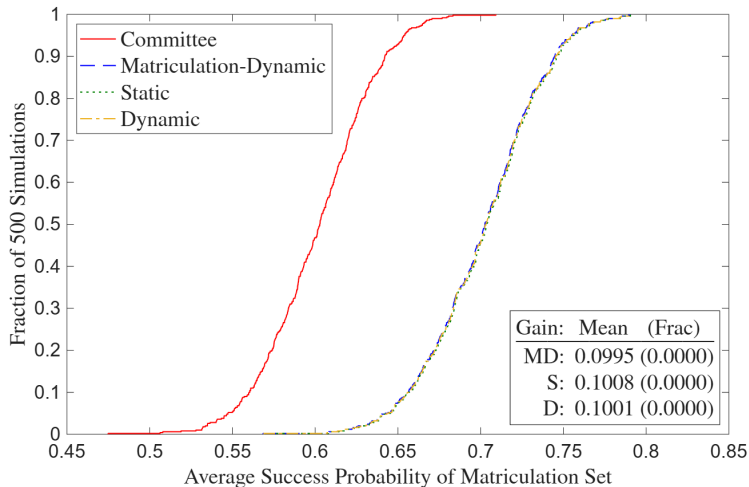
A4: Uncorrected Deviation Gains (Suboptimal)



A4: Uncorrected Matriculation Deviation Gains (Optimal)



A4: Uncorrected Matriculation Deviation Gains (Suboptimal)



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